Carbon Tariffs in Practice:

A Structural Gravity Analysis of the European Carbon Border Adjustment Mechanis

Bachelor Thesis

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Abstract

Preventing carbon leakage is a key challenge for effective climate action. The new Carbon Border Adjustment Mechanism will be introduced gradually, superseding the current free allocation system. To understand the interactions and effects during the transition period (2026-2034), I extend a structural gravity model to closely mirror EU climate policy using EU-ETS data. In a sequence of shocks, I introduce carbon pricing, free allocation and subsequently CBAM and free allocation fade out combined with the EU-ETS emission target. First results point to successful carbon leakage protection under this policy regime. Overall, this thesis provides a methodological framework for the analysis of EU climate policy which can be adapted for further research.

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List of Abbreviations

BACI	Bilateral Trade Database at the Product-Level
CBAM	Carbon Border Adjustment Mechanism
CEFS	Constant Elasticity of Fossil Fuel Supply
CEPII	Centre d'Études Prospectives et d'Informations Internationales
CES	Constant Elasticity of Substitution
CGE	Computable General Equilibrium
DEHSt	Deutsche Emissionshandelsstelle
EEA	European Economic Area
ETS	Emissions Trading System
EU	European Union
EU-ETS	European Emissions Trading System
EUTL	European Union Transaction Log
GDP	Gross Domestic Product
GHG	Greenhouse Gas
GTAP	Global Trade Analysis Project
HS6	Harmonized System 6-digit
MLE	Maximum Likelihood Estimator
OLS	Ordinary Least Squares
PPML	Poisson Pseudo Maximum Likelihood
RTA	Regional Trade Agreement
UK	United Kingdom
US	United States
USD	United States Dollar

1 Introduction

Anthropogenic climate change is one of the greatest threats to our societies. Acknowledging this, the EU agreed on ambitious climate targets and policies with the "Fit for 55 Package", the "European Green Deal" and further policy programs. They set the target to become climate neutral by 2050, reduce emissions by 55% compared against 1990 levels by 2030, and most recently set the goal of a 90% reduction for 2040 (Erbach and Jensen, 2024; European Commission, 2024b). However, the stringency and success of these policies critically hinges on effective protection against carbon leakage (emissions shifting to regions with weaker regulations) and protection of European competitiveness. Both undermine the fight against climate change and the public support for these policies.

In 2026, the EU will start to introduce a new instrument to minimize carbon leakage and maintain competitiveness: the Carbon Border Adjustment Mechanism (CBAM). It will gradually supersede the current system of free allocation. Under this framework, firms receive a share of the certificates for free, depending on product specific efficient emission benchmarks. CBAM instead charges the same price producers face in the EU, to firms exporting to the EU. Since the success of these policies is crucial for achieving climate goals, there is a need for comprehensive understanding of their effects and interactions. While carbon border adjustments have been explored in quantitative trade models, there are only limited insights of how free allocation and CBAM will interact during the transition period (2026-2034).

This thesis addresses this gap by expanding the structural gravity model of Larch and Wanner (2024) to analyze the combined effects of these instruments on carbon leakage and European competitiveness. I implement a system similar to the EU-ETS with a unified carbon price and free allocation and in a second step introduce CBAM while free allocation is faded out. The focus of this thesis is the adaptation of the model. Preliminary results show successful carbon leakage protection under CBAM and positive effects on European competitiveness.

The remainder of the thesis proceeds as follows. In Section 2, I give an overview of the existing literature that focuses on carbon leakage in computable general equilibrium (CGE) models. Section 3 explains the EU climate policy setup while Section 4 is the main part of the thesis. It explains the model, derives new instruments and features, and sheds light on the methodology. Section 5 specifies the modeling approach, and Section 6

contains the description of the baseline data used for the model and of the data underlying the new parameters introduced. Section 7 presents some preliminary results taking the thesis beyond purely theoretical work. Finally, Section 8 concludes.

2 Literature Review

Climate policy in quantitative trade models is well established in economic research. In this review, I will focus on the literature most closely related to the subject of this thesis and its methodology.

An important paper introducing an energy sector in an Eaton and Kortum (2002)-type equilibrium gravity model is Egger and Nigai (2015). They analyze how energy taxes affect demand, emissions, and welfare. The research by Duan et al. (2021), which tests the pollution haven hypothesis, is more directly concerned with carbon leakage. In a CGE framework with environmental regulation, they find no evidence that emission intensive industries tend to concentrate in countries with lax environmental regulation after reducing trade barriers. According to their analysis, environmental regulation only has a limited impact on international specialization, especially compared to productivity and trade costs. Farrokhi and Lashkaripour (2022) try to derive the optimal policy against free-riding and carbon leakage in a general equilibrium model framework. They conclude that non-contingent carbon border taxes, even under efficient design, fail to effectively limit global emissions. However, their analysis shows tariffs used as contingent penalties against free-riding to be highly effective in a climate club framework proposed by Nordhaus (2015). Artuc and Sommer (2024) present the important feature of within-firm carbon leakage in a general equilibrium trade model. When subject to a carbon tax, firms face the discrete decision of abating domestically or outsourcing intermediate production to another country. Introducing a border adjustment still leads to carbon leakage, but manages to increase abatement in both the country introducing the border adjustment and the country facing it. Finally, Sogalla (2024) specifically investigates EU climate policy in a quantitative general equilibrium trade model with heterogeneous firms. He compares the leakage protection instruments of free allocation and CBAM. Under free allocation income losses are lower while under CBAM carbon leakage protection works more effectively.

The strand of literature most important for this thesis begins with the paper by Larch and Wanner (2017) developing a structural gravity model for the quantitative analysis and decomposition of emission changes following the introduction of carbon tariffs. They show that carbon tariffs lead to lower leakage rates by also lowering trade flows and welfare. The model builds the foundation for subsequent papers analyzing different policies. Korpar et al. (2023) conduct an analysis of the first proposal for a carbon border adjustment mechanism by the EU Commission in 2021. They fully focus on the carbon tariff excluding the free allocation feature and the parallel reduction of the EU-ETS cap finding only limited effects on global emissions and even increasing emissions in the EU due to increased domestic production. Larch and Wanner (2024) adapt the model to evaluate the effects of withdrawals from the Paris Agreement. Furthermore, Richter and Wanner (2024) analyze various international climate policy options including tariffs imposed by single countries, the EU, or larger climate clubs. Since they do not specifically focus on CBAM, they use a simple modeling approach, neglecting various features of the coming EU regulation.

This shows that, while carbon border adjustment has been explored in the literature, an analysis of the interaction between free allocation and carbon border adjustment is still missing.

3 Policy Context

The goal of this thesis is to develop a model closely mirroring EU climate policy. The focus is on the European Emissions Trading System (EU-ETS), the key EU climate policy instrument, and policy tools connected to it. The EU-ETS is a Cap and Trade system covering roughly 40% of greenhouse gas (GHG) emissions in the EU (Erbach and Foukalová, 2023). Polluters in sectors covered by the EU-ETS have to surrender certificates for each ton of GHG they emit in a year. These certificates can be bought in auctions or on secondary markets. They are also partially obtained under free allocation. By linearly lowering the cap, the system aims to reduce emissions by 62% compared to 2005 emission levels in the covered sectors by 2030 (Erbach and Foukalová, 2023). The uniform certificate price for all polluters adjusts on the market, leading to an efficient reduction path. Industries in the following sectors fall under EU-ETS legislation: Energy (electricity and heat generation), Metals (iron, steel, aluminum), Mineral Industries (cement, lime,

glass, ceramics), Chemicals (ammonia, hydrogen, nitric acid), Pulp and Paper (paper, cardboard, pulp), Aviation, and Maritime Transport (European Parliament and Council, 2003). The EU-ETS covers the entire European Economic Area (EEA). To prevent carbon leakage and maintain global competitiveness, firms receive free certificates based on a system of benchmarks. These are defined as the tons of emissions per ton of the product emitted by the 10% most efficient production sites in the EU. Historic activity levels are used to allocate the amount of certificates to each factory. Only certificates for emissions exceeding these benchmarks must be bought at market prices. The certificates can also be sold. Solely the energy sector is exempted from this rule, because energy production faces no carbon leakage risk (Stoefs, 2022). The free allocation system builds on a quantity (tons of product) based framework which is not directly compatible with the model described in Section 4. The value of the certificates obtained for free can be expressed as:

$$\text{Value Free Allowance} = \text{benchmark} \cdot \text{historic activity level} \cdot \text{allowance price}$$

This system will be superseded gradually by CBAM starting with its introduction in 2026. CBAM is another carbon leakage protection and competitiveness instrument creating a level playing field between EU-based production and EU imports. Under this system, firms exporting to the EU pay the current EU-ETS price for the carbon content of their products. CBAM only covers the following products: electricity, cement and lime, basic iron and steel, aluminum, fertilizer, and hydrogen (DEHSt, 2025). The instrument will phase in while free allocation is phased out with the fade out factors shown in the timetable below (European Parliament and Council of the European Union, 2023).

Table 1: Gradual phase-out of free allocation for CBAM goods

Year	2026	2027	2028	2029	2030	2031	2032	2033	2034
Fade Out Factor	97.5%	95%	90%	77.5%	51.5%	39%	26.5%	14%	0%

Notes: The table shows the progression of free allocation fade out factors for CBAM goods from 2026 to 2034, with complete fade out by 2034 (European Parliament and Council of the European Union, 2023).

During this transitional period, emissions embedded in the imported goods are adjusted for the amount of free allocation which the same good receives in the EU. This

yields "covered emissions" which then face the EU-ETS price as a de-facto tariff (European Commission, 2025).

The "covered emissions intensity" of a product is obtained by deducting the free allocation benchmark emission intensity multiplied by the free allocation fade out factor from the imported goods emission intensity. Again, this builds on a quantity-based framework.

$$\text{Covered Emission Intensity} = \text{imports emission intensity} - \text{benchmark} \cdot \text{fade out}$$

The EU regulation intends to also adjust the payment for the carbon price "effectively paid in the country of origin" (European Commission, 2024a). However, the rules for this process are not yet published at the time of writing.

4 The Model

In this section, I describe the structural gravity model and the climate policy instruments in it. The multi-sector, multi-factor model with constant elasticity of fossil fuel supply (CEFS) builds on Larch and Wanner (2017, 2024), with extensions from Richter and Wanner (2024). In the model, emissions are proportional to the use of an internationally tradable fossil resource. The fossil resource is included in the energy production function, whereas energy is an input in the multi-factor Cobb-Douglas production function. Carbon pricing and carbon tariffs can change emissions, production, and trade patterns. The CEFS function creates a second carbon leakage channel based on the fossil resource price next to the trade-based channel of carbon leakage.

4.1 Structural Gravity Models

The thesis builds on counterfactual analysis with structural gravity models. Gravity models are a class of quantitative equilibrium trade models based on the idea that, just like gravity, bilateral trade depends positively on mass or size (GDP) and negatively on distance. In structural gravity models, this basic intuition is underpinned by microeconomic foundations on the supply and demand side grounding the model theoretically (Head and Mayer, 2014). There are different types of structural gravity model frameworks like the Armington model, where countries produce differentiated goods and consumers have

constant elasiticty of substitution (CES) preferences (Anderson and van Wincoop, 2003) or monopolistic competition following Krugman et al. (1980). The model by Larch and Wanner (2024) uses an Eaton and Kortum (2002)-type gravity equation based on a Ricardian model of international trade with perfect competition and heterogeneous technology. For the baseline calibration, bilateral trade costs were estimated from observed bilateral trade flows using a Poisson Pseudo Maximum Likelihood (PPML) estimator (Larch and Wanner, 2024). The counterfactual analysis proceeds by exogenously changing policy variables, solving for the new general equilibrium, and comparing the resulting trade flows, incomes, or emissions to the baseline scenario.

4.2 Supply

4.2.1 Goods Production

Each country in the set of countries ($j \in \mathcal{N}$) produces in each sector in the set of sectors ($l \in \mathcal{L}$) according to the Cobb-Douglas goods production function:

$$q_l^i = A_l^i \left(E_l^i\right)^{\alpha_{lE}^i} \prod_{f \in \mathcal{F}} \left(V_{lf}^i\right)^{\alpha_{lf}^i} \tag{1}$$

Here, A_l^i is the productivity parameter. α_{lE}^i and α_{lf}^i are the Cobb-Douglas production cost shares. V_{lf}^i denotes the input of production factor $f \in \mathcal{F}$ whereas E_l^i is the energy usage in production. Each country's factor endowment V_f^i is fixed. Factors are mobile across sectors but immobile across countries. Markets operate under perfect competition. Subsequently, goods are priced at marginal costs:

$$p_l^i = \frac{\Gamma_l^i}{A_l^i} \left(e_l^i\right)^{\alpha_{lE}^i} \prod_{f \in \mathcal{F}} \left(w_f^i\right)^{\alpha_{lf}^i} \tag{2}$$

where $\Gamma_l^i = \left(\alpha_{lE}^i\right)^{-\alpha_{lE}^i} \prod_{f \in \mathcal{F}} \left(\alpha_{lf}^i\right)^{-\alpha_{lf}^i}$ is the energy price and w_f^i are the factor prices in country i.

4.2.2 Energy Production

Energy is the only factor not endowed by countries and must be produced according to the Cobb-Douglas energy production function:

$$E^i = A_E^i \left(R^i\right)^{\xi_R^i} \prod_{f \in \mathcal{F}} \left(V_{Ef}^i\right)^{\xi_f^i} \tag{3}$$

The input cost shares of the factors and the fossil resource are denoted by ξ_f^i and ξ_R^i while the usage of the fossil resource is defined by R^i. This resource is traded frictionlessly at zero costs, leading to a perfectly integrated world market with a single fossil resource price r. Next to the resource price, the energy price is defined by factor prices and the technology parameter:

$$e_l^i = \frac{\Gamma_E^i}{A_E^i} \left((1 + \lambda_l^i) r\right)^{\xi_R^i} \prod_{f \in \mathcal{F}} \left(w_f^i\right)^{\xi_f^i} \tag{4}$$

where $\Gamma_E^i = \left(\xi_R^i\right)^{-\xi_R^i} \prod_{f \in \mathcal{F}} \left(\xi_f^i\right)^{-\xi_f^i}$ and λ_l^i is the carbon tax rate. The sectoral carbon tax requires deviating from the original model by Larch and Wanner (2024) by defining a sector-specific energy price. While energy production remains unified, the tax acts as a mark-up on energy production costs, changing the effective price of energy for carbon priced sectors. CO_2 emissions from production are proportional to the use of the fossil resource.

4.2.3 Fossil resource supply

For the global supply of the fossil resource R^W, the model by Larch and Wanner (2024) follows Boeters and Bollen (2012) using a CEFS function with supply shifter ζ:

$$R^W = \zeta \left(\frac{r}{P}\right)^{\eta} \tag{5}$$

Here, η is the supply elasticity and P is a world price index which can be decomposed into a national (P^i) and a country-sector level (P_l^i) price index. They relate to each other with $P \equiv \prod_{i \in \mathcal{N}} \left(P^i / \omega^i\right)^{\omega^i}$ and $P^i \equiv \prod_{l \in \mathcal{L}} \left(P_l^i / \gamma_l^i\right)^{\gamma_l^i}$ where γ_l^j expresses country j's expenditure share for sector l and ω_i represents each country's fossil resource endowment share ($\sum_{i \in \mathcal{N}} \omega^i = 1$). Countries contribute to the supply according to their endowment shares. Fossil resource supply responds to price changes in r with constant elasticity η, where higher elasticity reduces carbon leakage through energy prices.

4.3 Demand

4.3.1 Utility

For the utility function, I use the model specification in Richter and Wanner (2024), combining CES utility across country varieties and Cobb-Douglas utility across sectors.

$$U^j = \left[\prod_{l \in L} \left(\sum_{i \in N} \left(\beta_l^i\right)^{\frac{1-\sigma_l}{\sigma_l}} \left(q_l^{ij}\right)^{\frac{\sigma_l - 1}{\sigma_l}} \right)^{\frac{\sigma_l}{\sigma_l - 1}} \right]^{\gamma_l^j} \tag{6}$$

The consumers utility in country j depends on the preference parameter β_l^i, the amount of consumption from country i (q_l^{ij}), the elasticity of substitution σ_l between sector l varieties across countries, and the Cobb-Douglas expenditure share γ_l^i.

4.3.2 Gravity

The Eaton and Kortum (2002)-type gravity equation defines bilateral trade shares by sector. The model in Larch and Wanner (2024) uses iceberg trade costs T_l^{ij}, where $T_l^{ij} = T_l^{ji} \geq 1$ and $T_l^{ii} = 1$. The trade share is determined by the relative costs of goods from country i compared to the average cost of goods from all other countries in market j.

$$\pi_l^{ij} l = \frac{\left(\beta_l^i p_l^i T_l^{ij} \frac{\tau_l^{ij}}{\chi_l^i}\right)^{1-\sigma_l}}{\sum_{k \in \mathcal{N}} \left(\beta_l^k p_l^k T_l^{kj} \frac{\tau_l^{kj}}{\chi_l^k}\right)^{1-\sigma_l}} = \left(\frac{\beta_l^i p_l^i T_l^{ij} \frac{\tau_l^{ij}}{\chi_l^i}}{P_l^j} \right)^{1-\sigma_l} \tag{7}$$

For my model specification, I add an export subsidy, modeling free allocation of emission certificates to the equation (see section 4.4.3). The subsidy reduces import costs in country j for goods from country i.

4.4 EU climate policy instruments

In this subsection, I derive, and explain the policy instruments which constitute the key modifications I apply to the model by Larch and Wanner (2024).

4.4.1 CO_2 Emissions in the model

Emissions are proportional to the use of the fossil resource R^W. The emissions from sector l in country i can be derived from the expenditures for the fossil resource. ξ_R^i is

the fossil resource cost share in energy production. α^i_{lE} is the cost share of energy in goods production. The expenditure on the fossil resource by sector l is thus $\xi^i_R \alpha^i_{lE} Y^i_l$. Dividing the expenditure by the fossil resource price yields the level of emissions in sector l: $\frac{\xi^i_R \alpha^i_{lE} Y^i_l}{(1+\lambda^i_l)r}$. The models climate policy consists of lump-sum taxes, subsidies, and tariffs, requiring emission intensities defined as: $\frac{\xi^i_R \alpha^i_{lE} Y^i_l}{(1+\lambda^i_l)r} \frac{1}{Y^i_l} = \frac{\xi^i_R \alpha^i_{lE}}{(1+\lambda^i_l)r}$.

4.4.2 Carbon Pricing and Reduction Targets

The carbon tax remains similar to the original model. The model uses a lump-sum tax applied to the fossil resource price r. The tax rate is defined endogenously by the definition of a reduction target $\bar{R}^{EU}$. Hence, like in a Cap and Trade system, the total amount of emissions follows a predefined reduction path, adjusting endogenously for fossil resource consumption and price changes similar to certificates under EU-ETS.

$$\lambda^i_l = \begin{cases} 0 & \text{if } i \notin \text{EU} \vee l \notin ets, \\ \frac{\sum_{j \in \text{EU}} \xi^j_R \sum_{l \in ets} \alpha^j_{lE} Y^j_l}{\bar{R}^{\text{EU}} r} - 1 & \text{if } i \in \text{EU} \wedge l \in ets. \end{cases} \tag{8}$$

The tax rate equals current fossil resource expenditure $\sum_{j \in \text{EU}} \xi^j_R \sum_{l \in ets} \alpha^j_{lE} Y^j_l$ divided by target expenditure, where target expenditure is the quantity $\bar{R}^{EU}$ times the price r. The definition of countries ($EU \subset \mathcal{N}$) and sectors ($ets \subset \mathcal{L}$) subject to the pricing regime is different compared to the model by Larch and Wanner (2024). EU includes the countries of the EEA. Furthermore, I include Switzerland since, starting in 2020, the Swiss ETS is tied to the EU-ETS (Publications Office of the European Union, 2017). Moreover, I add the United Kingdom (UK) to the EU group in the model. The UK is included for several reasons: It is the only country whose emission pricing system has similar stringency and similarly strict rules as the EU-ETS (World Bank Group, 2023). In May 2025, the UK government and the European Commission signed a common understanding to start negotiations about linking UK-ETS and EU-ETS (Cabinet Office, 2025). Finally, the UK plans to introduce a CBAM in 2027 (Department for Energy Security & Net Zero, 2023). Unifying UK-ETS and EU-ETS makes deduction of domestic emission prices obsolete for UK exports to the EU. In my adaptation of the model, only some sectors are in the scope of this carbon tax[1]. This leads to a sectoral carbon tax in contrast to the original model.

[1]For details, see 4.4.5

4.4.3 Free Allocation Subsidy

To model free allocation, I follow Sogalla (2024) treating free allocation as a lump-sum subsidy.

With free allocation, firms only pay directly for a fraction of their emissions. However, opportunity costs from free certificates still incentivize emission reduction below the benchmark, converting free certificates into revenue streams. The subsidy I use in the model captures this dynamic. All emissions are fully covered by the tax, but an indirect income stream compensates for part of the tax costs by lowering the prices for consumers. This maintains the full incentive to decarbonize, but lowers carbon leakage by improving global competitiveness.

$$\chi_l^i = \begin{cases} 1 & \text{if } i \notin \text{EU} \vee l \notin ets, \\ 1 + \frac{\xi_R^i \alpha_{lE}^i}{(1+\bar{\lambda}_l^i)\bar{r}} \cdot s_l^i \cdot o_l^i \cdot \lambda_l^i r & \text{if } i \in \text{EU} \wedge l \in ets. \end{cases} \tag{9}$$

The lump-sum subsidy rate lowers consumer prices by the value of free allocation. This value depends on the carbon tax ($\lambda_l^i r$) and free allocation quantity. The latter is defined by the free allocation share s_l^i, which I derive from empirical free allocation data from the EU-ETS described in 6.2. This share is equivalent to free allocation benchmarks of the EU and creates a bridge between the quantity-based framework of the EU regulation and the cost share framework of the model. $\frac{\xi_R^i \alpha_{lE}^i}{(1+\bar{\lambda}_l^i)\bar{r}}$ is the emission intensity in sector l of country i of which the respective share s_l^i is subsidized. Appendix A proves the equivalence to the EU system.

Using shares instead of benchmarks does not fully reward cleaner production. If the emission intensity decreases due to a higher denominator following the carbon tax introduction, multiplying the intensity with the free allocation share also reduces subsidy payment. In a benchmark-based system, free allocation is constant independent of the firms decarbonisation progress.

To avoid this disincentive, I set the carbon tax and the fossil resource price in the emission intensity to the pre-shock level indicated by $\bar{\lambda}_l^i$ and $\bar{r}$ fixing these baseline intensities that determine the magnitude of the subsidy.

The carbon tax is implemented directly at the energy production level increasing the fossil resource price r. However, the subsidy targets the sector specific carbon content and must thus be implemented at the product level.

The implementation through the gravity equation lowers consumer prices, improving global competitiveness as shown in equation (7):

$$\pi_l^{ij} = \frac{\left(\beta_l^i p_l^i T_l^{ij} \frac{\tau_l^{ij}}{\chi_l^i}\right)^{1-\sigma_l}}{\sum_{k\in\mathcal{N}} \left(\beta_l^k p_l^k T_l^{kj} \frac{\tau_l^{kj}}{\chi_l^k}\right)^{1-\sigma_l}} = \left(\frac{\beta_l^i p_l^i T_l^{ij} \frac{\tau_l^{ij}}{\chi_l^i}}{P_l^j}\right)^{1-\sigma_l} \tag{7}$$

4.4.4 Carbon Border Adjustment Mechanism

For the CBAM tariff rate, I use the approach in Richter and Wanner (2024) and adding the adjustment for free allocation. The tariff applies to non-EU countries exporting to an EU country. It also only applies to a subset of sectors $cbam \subset \mathcal{L}$ covering fewer sectors than ets, consistent with EU regulation[2]. The tariff on goods from country i sold to country j in sector l is defined as:

$$\tau_l^{ij} = \begin{cases} 1 + t_l^{ij} & \text{if } i \notin \text{EU} \wedge j \in \text{EU} \wedge l \in \text{cbam}, \\ 1 & \text{if } i \in \text{EU} \vee j \notin \text{EU} \vee l \notin \text{cbam}. \end{cases} \tag{10}$$

The tariff rate imposed is:

$$t_l^{ij} = max\left\{\left(\frac{\xi_R^i \alpha_{lE}^i}{r} - \frac{\xi_R^j \alpha_{lE}^j}{(1+\bar{\lambda}_l^j)\bar{r}} \cdot s_l^j \cdot o_l^j \cdot \bar{p}_l^{ji}\right) \cdot \lambda_l^j r, 0\right\} \tag{11}$$

In its basic structure, the tariff rate is determined by the carbon intensity $\frac{\xi_R^i \alpha_{lE}^i}{r}$ in the exporting country multiplied with the price of emissions in the importing country $\lambda_l^j r$. However, due to free allocation in the importing country, a part of the exporter's emission intensity is exempt from the tariff. For the tariff rate, the subsidized emission amount per product value $\frac{\xi_R^j \alpha_{lE}^j}{(1+\bar{\lambda}_l^j)\bar{r}} \cdot s_l^j \cdot o_l^j$ is deducted from the exporter's emission intensity. $\bar{p}_l^{ji} \equiv \bar{p}_l^j / \bar{p}_l^i$ corrects for price differences between countries. Appendix A proves the equivalence to the EU system.

Since emissions depend on product quantity rather than value, identical cost shares can represent different emission levels when factor prices and subsequently final product prices vary. The price ratio harmonizes these intensities for accurate comparison. In Appendix B, I describe two example scenarios showing the bias introduced by directly comparing emission intensities defined relative to the product value and how the price

[2]For details, see 4.4.5

correction factor solves this. The tariff rate is limited to 0 to forestall negative rates.

Based on data obtained from the State and Trends of Carbon Pricing Dashboard (World Bank Group, 2023), I evaluate carbon pricing schemes outside the EU. Since there is no detailed information on the adjustment of CBAM payments to domestic carbon prices, I assume that the EU will only accept pricing systems with similar stringency, scopes, and principles. Most countries use grandfathering or free allocation systems, making effective polluter payments unclear. They allow offsets (sometimes unlimited), cover only some fuels, permit various exemptions, or cover few industries. Some countries like Japan, Ukraine or Argentina follow similar principles as the EU-ETS but charge negligible prices below 2 USD. For a detailed break down, see Appendix C.

Based on this assessment, I omit the adjustment for domestic carbon prices and except for the United Kingdom, which I include in the EU-group (see 4.4.2).

4.4.5 Sector Coverage for EU-ETS and CBAM

While the EU defines products and industry branches covered by EU-ETS and CBAM, the model uses broad aggregate sectors. Therefore, sector coverage requires simplifications and approximations. Based on the list of products in the scope of EU-ETS (European Parliament and Council, 2003) and CBAM (European Parliament and Council of the European Union, 2023) as well as the composition of the 13 tradable model sectors following Richter and Wanner (2024), I define the sets of sectors *ets* and *cbam*.

EU-ETS product groups align closely with the metal, mineral, and paper sectors. Another group of EU-ETS covered products falls in the chemical sector in the model. However, this sector also consists of pharmaceutical products not in the scope of the EU-ETS. Nonetheless, I include the chemical sector to cover its the emission intensive production processes. In addition, aviation and maritime transport face EU-ETS prices. These fall within the model's service sector, which also encompasses a large group of products not under EU-ETS regulation. Hence, I exclude the service sector from the carbon tax. In the EU even fewer products fall under CBAM. However, since these products belong to the model sectors chemical, metal, and mineral I include the entirety of these sectors in *cbam*.

Energy falls under both EU-ETS and CBAM following EU regulation. Yet in the model, energy is an intermediate good that is produced but is used only domestically as

input for the production of final goods. Subsequently, it cannot be addressed by CBAM. The carbon tax utilizes the energy sector by charging the tax at the energy price level, leading to higher energy prices for EU-ETS covered sectors. Thus, it is partially covered in connection with the final goods sectors. At the same time, there is no free allocation subsidy lowering energy prices because energy is exempted from free allocation in the EU.

4.5 Income and Equilibrium Conditions

4.5.1 Income Equation

There are four income sources of a country: (1) the factor income from production factors, (2) the income generated from the fossil resource supply, (3) the net carbon tax income as the difference between carbon tax income and free allocation subsidy expenditure, and (4) the carbon tariff income:

$$Y^i = \sum_{f\in\mathcal{F}} I_f^i + I_R^i + \xi_R^i \sum_{l\in ets}\left(\frac{\lambda_l^i}{1+\lambda_l^i}\right)\alpha_{lE}^i Y_l^i(1-s_l^i o_l^i) + \sum_{l\in cbam}\sum_{j\in\mathcal{N}}(\tau_l^{ji}-1)\frac{\pi_l^{ji}}{\tau_l^{ji}}\gamma_l^i Y^i \quad (12)$$

The factor income can be expressed as $I_f^i \equiv w_f^i\left[V_{Ef}^i + \sum_{l\in\mathcal{L}} V_{lf}^i\right]$ and the fossil resource income equals $I_R^i \equiv \omega^i R^W r$. Furthermore, $Y_l^i \equiv q_l^i p_l^i$ is the production value in sector l of country i.

This equation is based on Richter and Wanner (2024) with the change to net tax income which can be decomposed to tax income and subsidy expenditure as follows:

The tax income equals emissions covered by the tax $\frac{\xi_R^i\alpha_{lE}^i Y_l^i}{(1+\lambda_l^i)r}$ multiplied by the price of emissions $\lambda_l^i r$.

$$\sum_{l\in ets}\frac{\xi_R^i\alpha_{lE}^i}{(1+\lambda_l^i)r}Y_l^i\lambda_l^i r = \xi_R^i\sum_{l\in ets}\left(\frac{\lambda_l^i}{1+\lambda_l^i}\right)\alpha_{lE}^i Y_l^i$$

The subsidy expenditure equals emissions covered by free allocation $\frac{\xi_R^i\alpha_{lE}^i}{(1+\lambda_l^i)r}Y_l^i s_l^i o_l^i$ multiplied by the price of emissions $\lambda_l^i r$.

$$\sum_{l\in ets}\left(\frac{\xi_R^i\alpha_{lE}^i}{(1+\lambda_l^i)r}Y_l^i s_l^i o_l^i\right)\lambda_l^i r = \xi_R^i\sum_{l\in ets}\left(\frac{\lambda_l^i}{1+\lambda_l^i}\right)\alpha_{lE}^i Y_l^i$$

4.5.2 Trade Balance, Market Clearing and Equilibrium

In the model, all markets, namely national factor and energy as well as international goods and fossil resource markets, are assumed to clear. Trade is balanced by assumption. These equilibrium conditions differ from the original model in Larch and Wanner (2024) and Richter and Wanner (2024) in one important aspect. The subsidy lowers consumer prices of the goods. If the income of country i is expressed as the spending of country j, this spending in j differs from what producers in i receive. Consumers in j only pay the subsidized price, while producers in i receive the full price. This requires correction by the lump-sum subsidy, analogous to the tariff adjustments. This changes the trade balance condition (13) and the goods market clearing condition (16).

$$\sum_{i\in\mathcal{N}}\sum_{l\in\mathcal{L}}\frac{\pi_l^{ij}\chi_l^i}{\tau_l^{ij}}\gamma_l^j Y^j = \sum_{j\in\mathcal{N}}\sum_{l\in\mathcal{L}}\frac{\pi_l^{ji}\chi_l^j}{\tau_l^{ji}}\gamma_l^i Y^i \tag{13}$$

$$E^i = \sum_{l\in\mathcal{L}} E_l^i \tag{14}$$

$$V_f^i = \sum_{l\in\mathcal{L}} V_{lf}^i + V_{Ef}^i \tag{15}$$

$$Y_l^i = \sum_{j\in\mathcal{N}}\frac{\pi_l^{ij}\chi_l^i}{\tau_l^{ij}}\gamma_l^j Y^j \tag{16}$$

$$R^W r = \sum_{i\in\mathcal{N}}\sum_{l\in\mathcal{L}}\frac{1}{1+\lambda_l^i}\xi_R^i\alpha_{lE}^i Y_l^i \tag{17}$$

Based on this I formulate a general definition of an equilibrium in this adapted model version:

Definition 1. For given factor endowments V_f^i, productivity parameters A_l^i and A_E^i, preference parameters β_l^i and σ_l, trade costs T_l^{ij}, free allocation shares s_l^i, price correction factors $\bar{p}_l^{ji}$, Cobb-Douglas production function shares α_f^i, α_{lE}^i, ξ_R^i, ξ_f^i, fossil resource supply parameters ζ and η, an equilibrium under climate policy structure EU, ets, $cbam$, $\bar{R}^{EU}$, o_l^i is a set of factor prices w_f^i, energy prices e_l^i, carbon taxes λ_l^i, free allocation subsidy χ_l^i, carbon tariffs τ_l^{ij} and carbon tariff rate t_l^{ij}, world fossil resource price r, global fossil resource supply R^W that satisfy equilibrium conditions (1)-(17)

4.5.3 Equilibrium in Exact Hat Notation

The model depends on several structural parameters including factor endowments V_f^i, productivity parameters A_l^i and A_E^i, and preference shifters β_l^i. Estimating the levels of these parameters would require strong identifying assumptions, as productivity and preferences are inherently difficult to measure empirically.

This identification challenge is addressed using the "exact hat algebra" approach, popularized by Dekle et al. (2007, 2008) and formalized in Costinot and Rodríguez-Clare (2014). By re-expressing the equilibrium system in terms of changes from a baseline equilibrium, many structural parameters cancel out of the equilibrium conditions and do not require separate identification for counterfactual analysis.

Therefore, the model is solved using the "exact hat notation". The following section rewrites the equilibrium in terms of changes of the variables expressed as $\hat{x} \equiv x'/x$ where x is the baseline value and x' the counterfactual value of the variable.

Definition 2. Let w_f^i, e_l^i, r, R^W be a baseline equilibrium under climate policy structure λ_l^i, χ_l^i, τ_l^{ij}, t_l^{ij} and $w_f^{i'}$, $e_l^{i'}$, r', $R^{W'}$ be a counterfactual equilibrium under climate policy structure EU, ets, $cbam$, $\bar{R}^{EU'}$, $o_l^{i'}$. Then, $\hat{w}_f^i$, $\hat{e}_l^i$, $\hat{r}$, $\hat{R}^W$, $\widehat{1+\lambda_l^i}$, $\hat{\chi}_l^i$, $\hat{\tau}_l^{ij}$, $\hat{t}_l^{ij}$ satisfy the following equilibrium conditions (18)-(29):

Carbon Tax Change:

$$\widehat{1+\lambda_l^i} = \begin{cases} 1 & \text{if } i \notin EU \vee l \notin ets, \\ \frac{\sum_{k\in EU}\xi_R^k \sum_{l\in ets}\alpha_{lE}^k \sum_j (\hat{\pi}_l^{kj}\pi_l^{kj}\hat{\chi}_l^k\chi_l^k/\hat{\tau}_l^{kj}\tau_l^{kj})\gamma_l^j Y^{j'}}{\sum_{k\in EU}\xi_R^k \sum_{l\in ets}\alpha_{lE}^k \sum_j (\pi_l^{kj}\chi_l^k/\tau_l^{kj})\gamma_l^j Y^j} \left(\frac{\bar{R}^{EU'}}{\bar{R}^{EU}}\hat{r}\right)^{-1} & \text{if } i \in EU \wedge l \in ets. \end{cases} \tag{18}$$

Free Allocation Subsidy Change:

$$\hat{\chi}_l^i = \begin{cases} 1 & \text{if } i \notin EU \vee l \notin ets, \\ \frac{1+\xi_R^i\alpha_{lE}^i \cdot s_l^i \cdot o_l^{i'} \cdot [\lambda_l^{i'}\hat{r}/(1+\lambda_l^i)]}{1+\xi_R^i\alpha_{lE}^i \cdot s_l^i \cdot o_l^i \cdot [\lambda_l^i r/(1+\bar{\lambda}_l^i)\bar{r}]} & \text{if } i \in EU \wedge l \in ets. \end{cases} \tag{19}$$

CBAM Tariff Change:

$$\hat{\tau}_l^{ij} = \begin{cases} 1+\hat{t}_l^{ij}t_l^{ij} & \text{if } i \notin \text{EU} \wedge j \in \text{EU} \wedge l \in \text{cbam}, \\ 1 & \text{if } i \in \text{EU} \vee j \notin \text{EU} \vee l \notin \text{cbam}. \end{cases} \tag{20}$$

CBAM Tariff Rate Change:

$$\hat{t}_l^{ij} = \frac{max\left\{\left(\frac{\xi_R^i \alpha_{lE}^i}{\hat{r}r} - \frac{\xi_R^j \alpha_{lE}^j}{(1+\lambda_l^j)r} \cdot s_l^j \cdot o_l^{j'} \cdot \bar{p}_l^{ji}\right) \cdot \lambda_l^{i'} \hat{r}r, 0\right\}}{max\left\{\left(\frac{\xi_R^i \alpha_{lE}^i}{r} - \frac{\xi_R^j \alpha_{lE}^j}{(1+\bar{\lambda}_l^j)\bar{r}} \cdot s_l^j \cdot o_l^j \cdot \bar{p}_l^{ji}\right) \cdot \lambda_l^j r, 0\right\}} \tag{21}$$

Cost Change:

$$\hat{c}_l^i = \left(\hat{e}_l^i\right)^{\alpha_{lE}^i} \prod_{f \in \mathcal{F}} \left(\hat{w}_f^i\right)^{\alpha_{lf}^i} \tag{22}$$

Trade Share Change:

$$\hat{\pi}_l^{ij} = \frac{\left(\frac{\hat{\tau}_l^{ij} \hat{c}_l^i}{\hat{\chi}_l^i}\right)^{1-\sigma_l}}{\sum_k \pi_l^{kj} \left(\frac{\hat{\tau}_l^{kj} \hat{c}_l^k}{\hat{\chi}_l^k}\right)^{1-\sigma_l}} \tag{23}$$

Price Index Change:

$$\hat{P}_l^j = \left[\sum_i \pi_l^{ij} \left(\frac{\hat{c}_l^i \hat{\tau}_l^{ij}}{\hat{\chi}_l^i}\right)^{1-\sigma_l}\right]^{\frac{1}{1-\sigma_l}} \tag{24}$$

Fossil Resource Supply Change:

$$\hat{R}^W = \left(\frac{\hat{r}}{\hat{P}}\right)^{\eta} = \left(\frac{\hat{r}}{\prod_{i \in \mathcal{N}} \left(\prod_{l \in \mathcal{L}} \left(\hat{P}_l^i\right)^{\gamma_l^i}\right)^{\omega^i}}\right)^{\eta} \tag{25}$$

Factor Price Change:

$$\hat{w}_f^i = \frac{\sum_l (\xi_f^i \alpha_{lE}^i + \alpha_{lf}^i) \sum_j \hat{\pi}_l^{ij} \pi_l^{ij} \frac{\hat{\chi}_l^i \chi_l^i}{\hat{\tau}_l^{ij} \tau_l^{ij}} \gamma_l^{j'} Y^{j'}}{I_f^i} \tag{26}$$

Energy Price Change:

$$\hat{e}_l^i = \left((1+\lambda_l^{i'})\hat{r}\right)^{\xi_R^i} \prod_{f \in \mathcal{F}} \left(\hat{w}_f^i\right)^{\xi_f^i} \tag{27}$$

Fossil Resource price Change:

$$\hat{r} = \frac{\sum_i \sum_l \left(\frac{1}{1+\lambda_l^i}\right) \xi_R^i \alpha_{lE}^i \sum_j \hat{\pi}_l^{ij} \pi_l^{ij} \frac{\hat{\chi}_l^i \chi_l^i}{\hat{\tau}_l^{ij} \tau_l^{ij}} \gamma_l^j Y^{j'}}{\sum_i \sum_l \left(\frac{1}{1+\lambda_l^i}\right) \xi_R^i \alpha_{lE}^i \sum_j \pi_l^{ij} \frac{\chi_l^i}{\tau_l^{ij}} \gamma_l^j Y^j} (\hat{R}^W)^{-1} \tag{28}$$

Counterfactual Income:

$$Y^{j'} = \sum_{f \in \mathcal{F}} \hat{w}_f^j I_f^j + \hat{R}^W \hat{r} I_R^j + \xi_R^j \sum_{l \in \mathcal{L}} \left(\frac{\lambda_l^{j'}}{1 + \lambda_l^{j'}} \right) \alpha_{lE}^j (1 - s_l^j o_l^{j'}) \sum_{i \in \mathcal{N}} \hat{\pi}_l^{ji} \pi_l^{ji} \frac{\hat{\chi}_l^j \chi_l^j}{\hat{\tau}_l^{ji} \tau_l^{ji}} \gamma_l^i Y^{i'} \quad (29)$$

$$+ \sum_{l \in \mathcal{L}} \sum_{i \in \mathcal{N}} (\hat{\tau}_l^{ij} \tau_l^{ij} - 1) \hat{\pi}_l^{ij} \pi_l^{ij} \frac{\hat{\chi}_l^i \chi_l^i}{\hat{\tau}_l^{ij} \tau_l^{ij}} \gamma_l^j Y^{j'}$$

5 Modeling Approach

The analysis focuses on the two shocks described in Section 3, namely (i) the introduction of CBAM and the corresponding free allocation fade out starting in 2026 and (ii) the emission reduction target of 62% against 2005 enforced by the linear reduction of the cap until 2030.

The model baseline uses 2017 data. Although EU-ETS launched in 2005, prices remained negligible until 2018, when they rose significantly, and are expected to exceed 100 EUR by 2030.

To investigate the isolated effect of the CBAM introduction and the further reduction of the cap until roughly 2030, I implement two consecutive shocks. Directly implementing carbon pricing, CBAM, lower free allocation and the -62% target would make it impossible to disentangle the effects and to evaluate the interaction of CBAM and free allocation in the transition period.

The first shock is supposed to move the production, trade and emissions structure in the model to the year 2025, before the CBAM shock. The second shock then models CBAM, free allocation fade out and the further reduction needed, to achieve -62% against 2005.

Between 2005 and 2017, emissions in the sectors covered by EU-ETS were reduced by 26% according to Carbon Market Watch (2021). I assume that there was no carbon price until 2017 which is reasonable due to the low price level in the EU-ETS before 2018. Furthermore, I assume that the baseline data reflects this emission reduction as well. Compared to 2005 levels, 2024 saw a reduction in emissions by 50% in EU-ETS sectors (European Commission, 2025). To close the gap between 2017 and 2024, I implement the first shock introducing the carbon tax and the free allocation subsidy to the model, with

a reduction target of -32%, because $50/74 \approx 0.68$. This serves as an intermediary step, creating the new baseline data, which is then used as the reference point for the second shock.

Building on this new setting, I introduce the carbon tariff, fade out free allocation with a fade out rate of $o_l^i = 0.5$ matching the fade out timeline of the EU. The change in these policy instruments goes along with an additional emission reduction by 24%, because $38/50 = 0.76$, to achieve the 62% reduction target set by the EU. This is the main shock of the analysis.

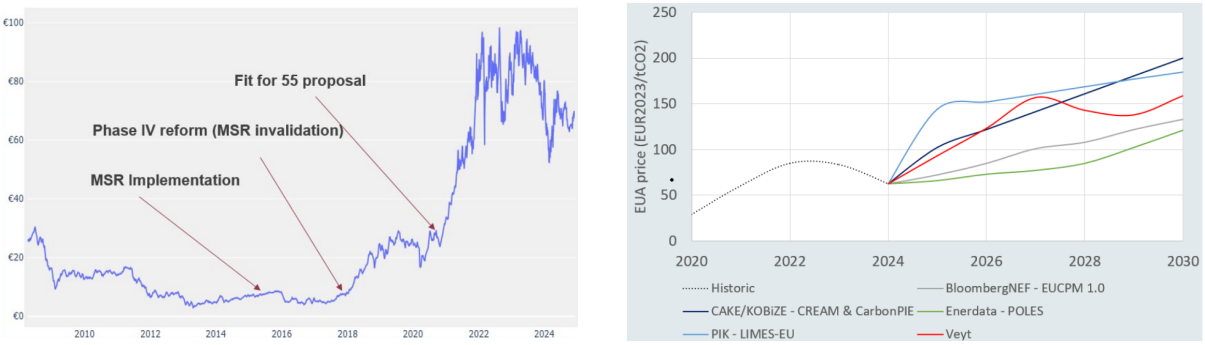


Figure 1: Historic price levels and price development for EU-ETS allowances

Notes: The left graphic shows historic prices for EU-ETS allowances based on data by https://sandbag.be/carbon-price-viewer. The right graphic shows price forecasts by different institutes based on different EU-ETS models from 2024. Both taken from Görlach et al. (2025).

6 Data

In this section, I explain the data used in the model, covering baseline data sources while focusing on data newly introduced to the model in the context of its adaptation.

6.1 Baseline Model Data and Parameters

I use the baseline data calibrated in Richter and Wanner (2024). The baseline data builds on data from the Global Trade Analysis Project (GTAP) 11 Database (Aguiar et al., 2022) where the most recent data from 2017 is used. It includes data on 159 countries (or aggregate regions) and 65 GTAP sectors which are then aggregated to 13 tradable sectors, a non-tradable sector, and an energy sector in the model. The production cost shares α_{lf}^i, α_{lE}^i, ξ_f^i, and ξ_R^i as well as the estimates for the sectoral elasticities of substitution σ_l also come from the GTAP database. Additional parameters for the gravity estimation come from CEPII (Head et al., 2010) and RTA database (Egger and Larch,

2008). Baseline bilateral trade shares, fitted using estimated bilateral trade costs, follow the gravity estimation from Larch and Wanner (2024). Furthermore, the authors use an "emission-weighted average of the fuel specific supply elasticities" (Richter and Wanner, 2024) which is based on Boeters and Bollen (2012) and yields a fossil supply elasticity of $\eta = 2.35$.

6.2 Free Allocation Data

The free allocation shares s_l^i estimate which fraction of emissions is covered by free allocation in each sector and country. Therefore, I use data from the European Union Transaction Log (EUTL) which tracks data related to the EU-ETS. I obtained the data from the EUETS.info database by Abrell (2024a). See Abrell (2024b) for detailed data descriptions and sourcing processes. The dataset provides information on the production facility level. For each facility, the dataset provides country, activity-id, number of free allocated certificates, and number of surrendered certificates from 2005 to 2023. Since free allocation rules changed over time, I only use data from the current ETS trading period IV, which started in 2021. Using the activity-id descriptions and the GTAP sector descriptions, as well as the sectoral aggregation concordance from Richter and Wanner (2024), I map facilities to model sectors. After analyzing the number of observations at the sector-country level for the 2021-2023 data, I cluster country groups as shown in Table 2 to ensure meaningful sample sizes (>30 observations) for each sector-country cluster combination.

The raw data includes many extreme allocation ratios, especially extremely high values. This can come from errors in the data, lower activity (because free allocation is based on historic activity levels) or from actual decarbonization. Distinguishing between these causes is difficult. Ideally, high ratios from successful decarbonization would be retained while excluding those from reporting errors or reduced production due to economic factors. I excluded observations above the 90th percentile because economic conditions during 2021-2023 likely reduced production in energy-intensive industries, artificially inflating free allocation ratios. After computing the aggregate free allocation ratios at the sector-country cluster level, I compare different filtering methods to ensure robustness of the results. Figure 2 compares the chosen 90th percentile filter to a 95th percentile cutoff. By the nature of this process, this leads to slightly higher values, but there are no

Table 2: Country Group Classification

Group	**Countries**
DACH+	Germany, Switzerland, Austria, Liechtenstein, Iceland
France	France
Iberian Peninsula	Spain, Portugal
NeLux	Netherlands, Luxembourg
Belgium	Belgium
Baltic Sea	Baltic States, Poland
Central Europe I	Hungary, Croatia, Slovenia
Central Europe II	Slovakia, Czech Republic
Mediterranean I	Malta, Italy
Mediterranean II	Greece, Bulgaria, Romania, Cyprus
North Atlantic	Ireland, United Kingdom

Notes: The clustering accounts for similar history, GDP p.c. and geographical proximity to ensure comparability. Iceland and Liechtenstein form an aggregate region in the GTAP data and thus need to have the same value. Therefore, Iceland is part of the DACH+ group.

irregular deviations in the empirical correlation and the slope is close to 1.

Free allocation ratios are computed by dividing total free allocation by total surrendered certificates within each sector-country cluster. The ratios show systematic differences by sector in both level and heterogeneity across countries, as shown in Figure 3. This implies differences in the carbon intensity by country and differences in benchmark stringency or recent activity levels by sector.

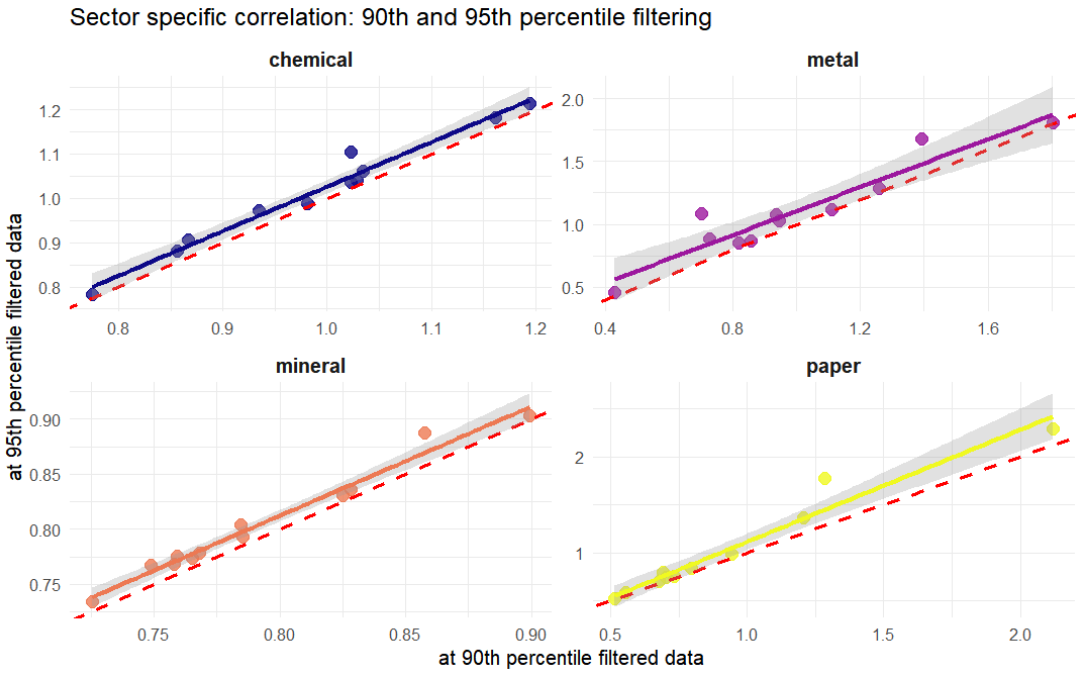


Figure 2: Sector specific correlation: 90th and 95th percentile filtering

Notes: The red line displays perfect correlation. The fitted lines show the correlation between the data resulting from the different filtering methods.

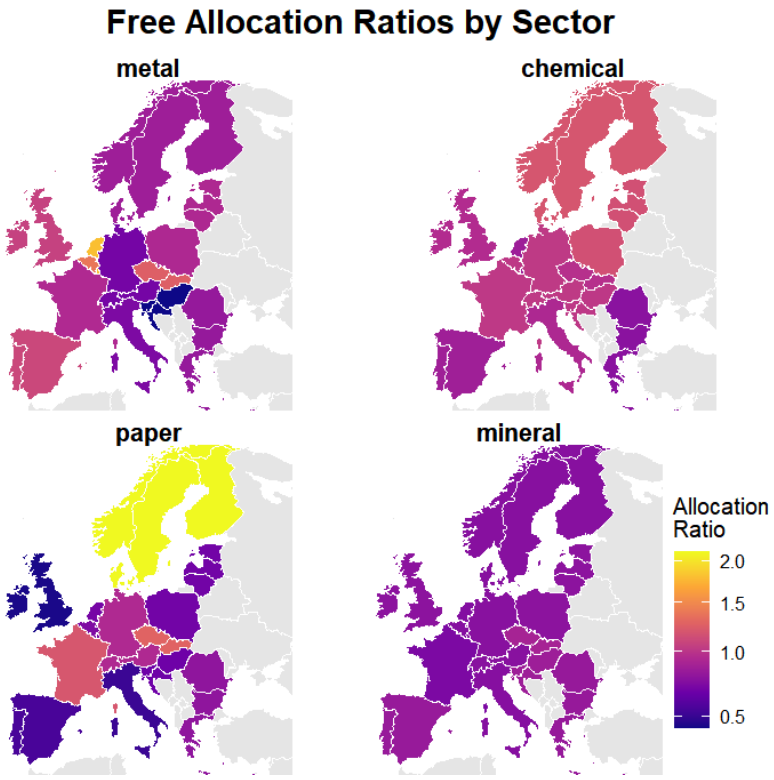


Figure 3: Map of Free Allocation Ratios by Sector

Notes: The level of the free allocation ratio is indicated by the common legend.

6.3 Price Ratio Data

As described in Section 4.4.4, the tariff rate needs a price correction factor to function properly. However, neither the baseline calibration process nor the "exact hat" approach yields values for price or cost levels. Therefore, I use exogenous prices obtained from the International Trade Database at the Product-Level (BACI) (Gaulier et al., 2008). This dataset, building on the United Nations Statistical Division's Comtrade database, contains bilateral trade flows between more than 200 countries for products at the HS6 nomenclature level. I compute product prices based on value (in USD) and quantity (in tons). Using concordance tables, I map HS6 codes to GTAP codes and model sectors (Decreux, 2021) and match GTAP country IDs with ISO3 codes (Baumann, 2023). Although I need country pair data eventually, I compute quantity-weighted price indices averaging over all observations of each exporting country in the respective model sector. This reduces extreme values through larger sample sizes and avoids missing country-pair data problems. Using these country-sector price indices, I compute country-pair and sector price ratios. Prices can only be positive but have no upper limit. The same is true for price ratios, where $p_l^i/p_l^j = k$ and $p_l^j/p_l^i = 1/k$ introduce reciprocal symmetry. Additionally, there are more than 25.000 observations for each sector. Thus, it is reasonable to assume a log-normal distribution for the price ratios which is approximately true based on the density plots and QQ-plots for each sector in Figures 4 and 5.

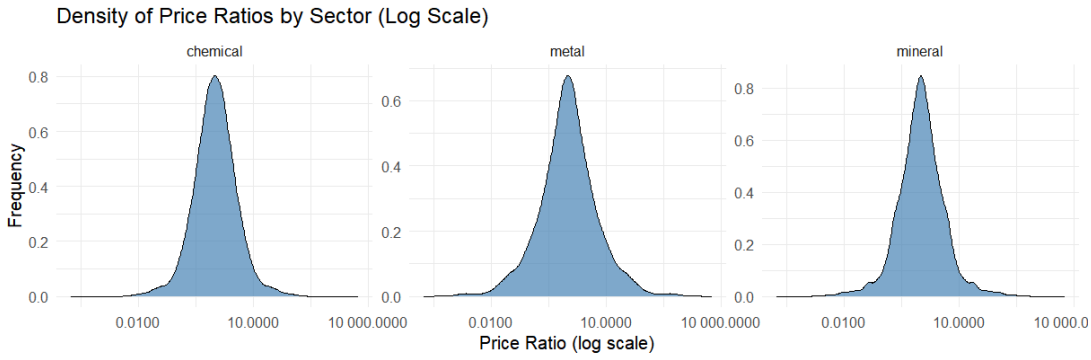


Figure 4: Density of Price Ratios by Sector (Log Scale)

Notes: Price ratios on $\log_{10}$ scale. The chemical sector most closely approximates log-normality, while metal and mineral sectors show discrete anomalies and small deviations from the theoretical log-normal benchmark.

Based on the distribution assumption, I cap extreme values at the 1.5 SD threshold for each sector. Table 3 shows the upper and lower threshold of the price ratios for 1.5 SD and 2 SD cutoffs and the percentage share of observations outside of these limits. As

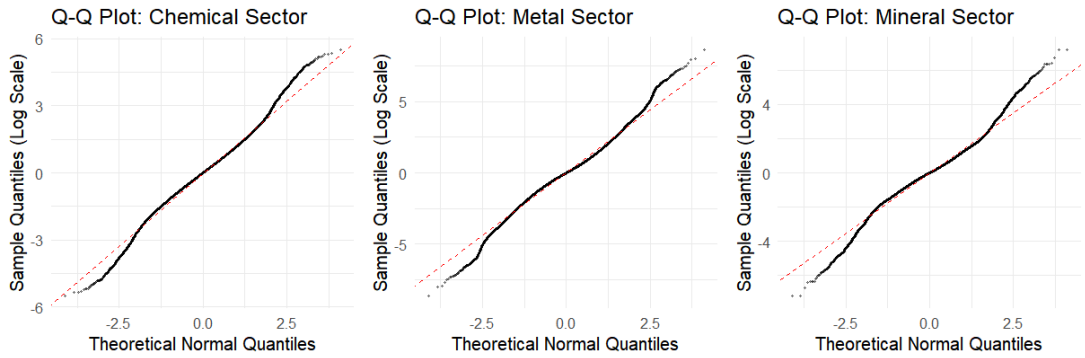


Figure 5: Q-Q Plot for Log-Normal Distribution by Sector

Notes: Price ratios were log-transformed and plotted against theoretical normal quantiles. Points following the diagonal line indicate that the original price ratios are log-normally distributed.

the precision of the price ratio values is not key to the model, but can have a large effect on the tariff rates, I choose the conservative thresholds which still affect at most 12.8% of observations. Additionally, values are capped rather than excluded, preserving their functional effect.

Table 3: Extreme value thresholds and affected observations by sector

Sector	Lower Threshold		Upper Threshold		% of Extreme Values	
	1.5 SD	2.0 SD	1.5 SD	2.0 SD	1.5 SD	2.0 SD
Mineral	0.12	0.06	8.22	16.60	10.81%	5.79%
Chemical	0.14	0.08	6.96	13.28	11.81%	5.33%
Metal	0.07	0.03	14.09	34.02	12.79%	5.87%

Notes: Thresholds are calculated based on standard deviations (SD) from the mean for each sector of the log-transformed data. "Lower Threshold" refers to values below the mean by 1.5 or 2.0 SD; "Upper Threshold" refers to values above the mean by 1.5 or 2.0 SD. "% Extreme" reports the share of observations that lie outside these thresholds, expressed as percentages.

7 Results and Discussion

This section briefly presents preliminary results to demonstrate model dynamics and functionality. However, these results require further validation through robustness checks, comparative scenarios, effect decomposition, and literature comparison.

7.1 First Shock: Carbon Tax and Free Allocation Subsidy

The first shock aligns baseline data with current EU policy by introducing a carbon tax mirroring EU-ETS with a 32% reduction target. Accompanying this shock is the corresponding free allocation subsidy (without fade out factor), lowering consumer prices

and carbon tax income. This serves as an intermediate step for the core analysis. Thus, I only present key results with further graphics and tables in Appendix D.

Table 4: Results First Shock

	Emission Reduction World	Emission Reduction EU	Emission Reduction EU (ETS sectors)	Leakage Rate	Real Income Change World
Results	-0.77%	-9.52%	-32%	31%	-0.026%

	Real Income Change EU	Real Income Change non-EU	Real Income Change resource rich	Carbon Tax Rate	Real Fossil Price Change
Results	-0.01%	-0.03%	-0.2%	48.09%	-0.33%

Based on the defined reduction target, emissions in EU-ETS sectors decrease by 32% leading to a 9.52% EU-wide reduction in emissions but only 0.77% globally (see Table 4). Emissions in all non-EU countries increase but at very low rates (see Figure 6). Real income marginally declines with a stronger decrease in resource-rich countries induced by the decrease in the fossil resource price. The carbon tax rate applied to this resource price for EU-ETS sectors is close to 50%. Most EU market shares in non-ETS sectors decrease marginally (around 0.1%), but increase modestly in carbon-priced sectors (0.03-0.41%). In follow-up work, analyzing net tax income would be valuable to compare the magnitude of tax and subsidy payments. While it seems that the subsidy can moderate the economic effects of the tax, its effectiveness in preventing carbon leakage is limited. With a leakage rate of 31%, roughly 1/3 of reductions in the EU is offset by an increase in other countries. Figure 7 shows that leakage shares are especially high in countries with a large industrial sector like the US, China, India, and Russia.

In summary, the first shock substantially changes European emissions with leakage effects on non-EU countries but only leads to minor economic changes. The equilibrium obtained after this shock is used as the new baseline for the second shock.

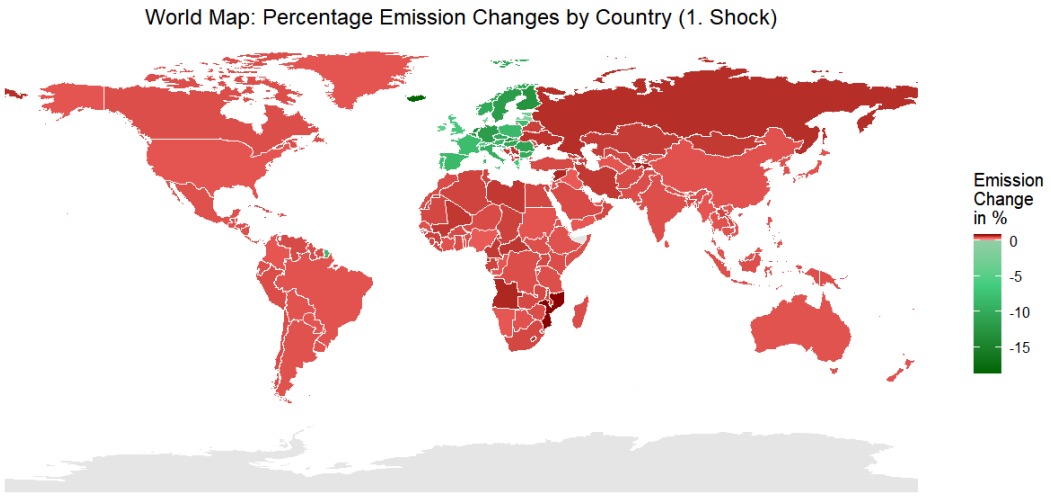


Figure 6: World Map: Emission Reduction by Country (1. Shock)

Notes: The color scale turns from green to red at 0. Changes in emissions are relative to the baseline scenario.

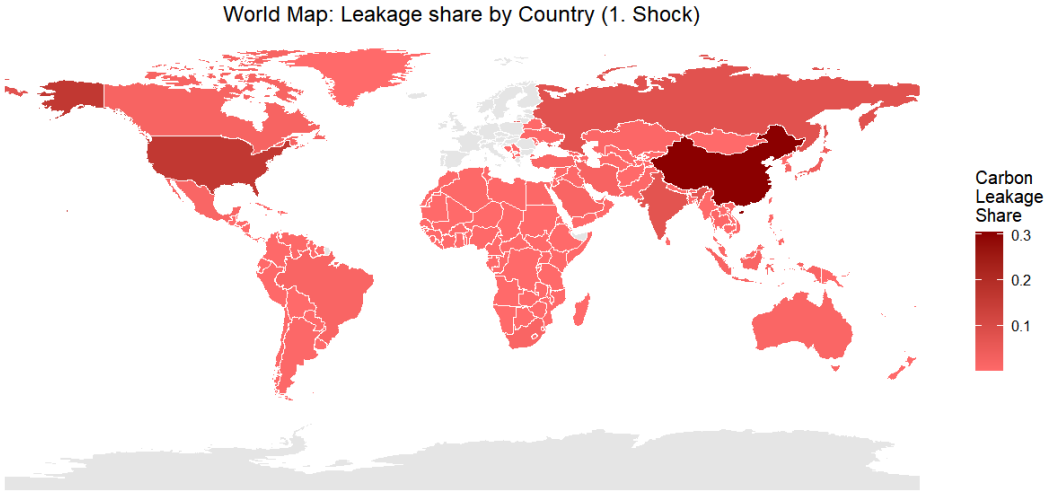


Figure 7: World Map: Carbon Leakage Share by Country (1. Shock)

Notes: The map shows how the amount of leakage emissions is attributable to the individual countries.

7.2 Second Shock: CBAM and Free Allocation Fade Out

The second shock describes the key policy change for this thesis. It introduces a carbon tariff mirroring CBAM while the free allocation subsidy is reduced by 50% for sectors covered by CBAM. Simultaneously, the carbon tax is increased to achieve an additional 24% reduction, which is a reduction compared to the original GTAP data by 50% and reaches the EU-ETS target of -62% against 2005 in 2030. The configuration of this shock roughly describes EU climate policy in 2030 compared to the status at the end of 2024.

Table 5: Results Second Shock

	Emission Reduction World	Emission Reduction EU	Emission Reduction EU (ETS sectors)	Leakage Rate	Real Income Change World
Results	-0.6%	-5.06%	-24%	-0.1%	-0.19%

	Real Income Change EU	Real Income Change non-EU	Real Income Change resource rich	Carbon Tax Rate	Real Fossil Price Change
Results	-0.03%	-0.2%	-0.26%	96.21%	-0.25%

By definition, the model yields a reduction in emissions for EU-ETS-covered sectors of 24%. This translates to an EU wide reduction of around 5% but only 0.6% worldwide (see Table 5). After the second shock, emission reductions extend beyond EU members to many countries worldwide, as shown in Figure 8.

This is also reflected in carbon leakage. The results of the second shock indicate that the absolute emission reduction worldwide is higher than in the EU, resulting in negative leakage emissions and a negative leakage rate of roughly 0.1%. This is a remarkable result and will be discussed at the end of this section. Figure 9 shows the distribution of positive and negative leakage. The negative total leakage emissions, underlying the computation of leakage shares, lead to shares above 1 or below -1 for some countries if the single countries absolute change in emissions is larger than the leakage emissions in absolute terms.

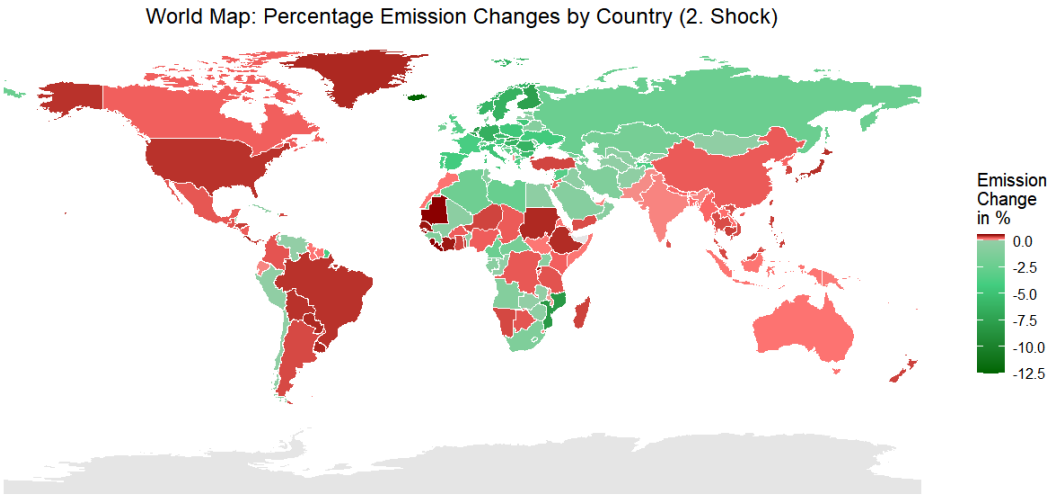


Figure 8: World Map: Emission Reduction by Country (2. Shock)

Notes: The color scale turns from green to red at 0. Changes in emissions are relative to the baseline scenario.

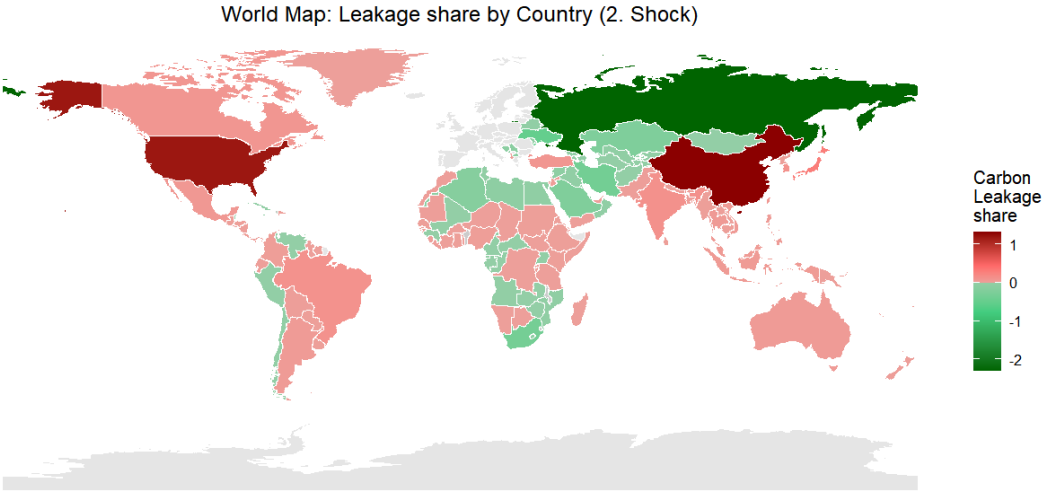


Figure 9: World Map: Carbon Leakage Share by Country (2. Shock)

Notes: The map shows how the amount of leakage emissions is attributable to the individual countries.

However, the economic results are less difficult to explain. Real income decreases worldwide by 0.19%, primarily in resource-rich countries. The EU only bears a small part of this loss. As displayed in Figure 10, some countries even gain in real income. Real wages change similarly, for details see Appendix E. Due to lower demand, the real fossil resource price decreases by 0.25%. The carbon tax rate nearly doubles compared to the first shock, now at 96.21%, making fossil resource prices in EU-ETS sectors nearly twice as high as in other sectors.

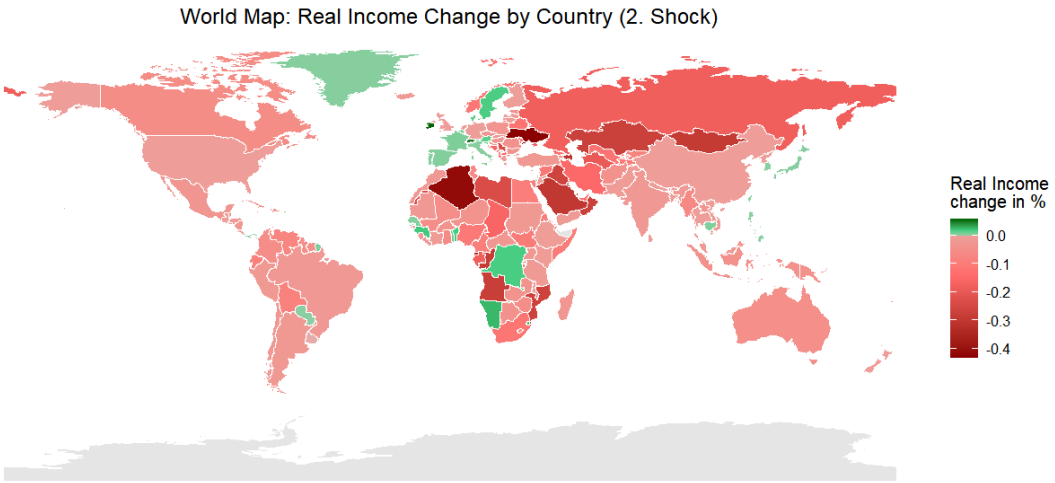


Figure 10: World Map: Real Income Change by Country (2. Shock)

Notes: The color scale turns from red to green at 0.

Table 6 shows the subsidy rates for selected countries with heterogeneity across countries and sectors depending on emission intensity and free allocation shares. Subsidy rates appear small relative to the tax rate, but they are applied to the final goods price, whereas the tax targets the fossil resource price directly. Compared to the first shock, the subsidy rates for CBAM sectors decreased due to the fade-out factor, while the rate applied to the paper sector increases with the increasing tax rate. For details, see Appendix E.

Tables 8 and 7 show the tariff rates imposed by Germany and Poland on selected trading partners. Countries with cleaner production, such as South Korea or the United States, face lower tariffs than, for example, India or Saudi Arabia.

Table 6: Subsidy Rate by Country and Sector (2. Shock)

Country	Chemical	Mineral	Metal	Paper
Germany	0.70%	0.41%	0.26%	0.95%
Spain	0.37%	0.63%	0.56%	0.49%
France	0.45%	0.38%	0.31%	0.64%
Italy	0.43%	0.41%	0.21%	0.55%
Poland	1.28%	1.04%	0.69%	1.08%

Notes: Selection of countries only for illustrative purposes.

Table 7: Poland: Imposed Tariff by Exporter and Sector

Sector	Brazil	China	India	South Korea	Saudi Arabia	Turkey	United States
Chemical	0%	4.5%	5.25%	3.54%	11.02%	0.27%	2.71%
Mineral	0%	3.65%	9.78%	0.85%	3.88%	4.05%	1.22%
Metal	1.38%	6.99%	9.7%	0.52%	2.26%	1.78%	1.3%

Notes: Tariff rate imposed by Poland to non-EU countries. Selection of trading partners only for illustrative purposes.

Table 8: Germany: Imposed Tariff by Exporter and Sector

Sector	Brazil	China	India	South Korea	Saudi Arabia	Turkey	United States
Chemical	0%	4.57%	5.29%	3.61%	11.17%	0.37%	2.76%
Mineral	0%	4.51%	10.88%	1.51%	4.7%	5.72%	1.92%
Metal	2.68%	7.62%	10.61%	1.27%	2.89%	2.7%	1.71%

Notes: Tariff rate imposed by Germany to non-EU countries. Selection of trading partners only for illustrative purposes.

The effect on global EU market shares is mixed, but small for sectors outside CBAM. The market shares in the mineral, chemical, and especially in the metals sector increase substantially. This is, based on the logic of CBAM pricing imports but not supporting exports, likely driven by increased domestic sourcing and intra-EU trade. These effects align with emission leakage pattern and trade effects discussed in more detail in the following.

Table 9: EU Market Share Changes by Sector (2. Shock)

	Agriculture	Apparel	Chemical	Equipment	Food	Machinery
Change in %	0.17	-0.63	0.59	-0.38	0.13	-0.87

	Metal	Mineral	Other	Paper	Service	Textile	Wood
Change in %	2.45	0.67	-0.48	0.15	0.21	-0.28	-0.03

Notes: Change in global market shares by sector of the aggregate EU region expressed as percentage points.

The negative carbon leakage rate raises questions about the model adaptation's validity especially compared to Richter and Wanner (2024) where the same basic model structure shows substantial positive leakage rates. Even though the details of the policy instruments differ, it is surprising to see completely diverging results. This requires further investigation and checks.

However, a first look at the results reveals coherent mechanisms driving the strong reduction in global emissions. Substantial emission reductions occur especially in the metal and chemical sector. I identify the non-EU countries with the largest absolute emission reduction in both sectors and analyze the tariff rates and the change in trade shares of these countries with the largest EU economies. Tables 10 and 11 show the 15 observations with the largest trade share reductions of the obtained samples. Extended tables including minerals appear in Appendix E. I also include the respective price ratios used in the tariff computation to rule out that very low price ratios (which are prone to data errors) distort the tariff rates. However, all observations have relatively high price ratios which reduce tariff rates.

The tables clearly demonstrate that high emission reductions go along with dramatic

falls in the trade shares with large EU countries driven by very high tariff rates. The high tariff rates are also a result of the high carbon tax. Although this is no conclusive evidence for the functioning of the model, it provides a first explanation. Sufficiently high tariff rates lead to dramatic drops in trade activity, lowering production and subsequently emissions of non-EU countries.

Table 10: Metal Sector: High Emission and Trade Share Reduction Exporters

Exporter	Importer	Trade Share Reduction (%)	Tariff Rate (%)	Price Ratio	Emission Reduction (kton)
Kazakhstan	Germany	-61.95	18.20	1.70	-3.49
Kazakhstan	Italy	-61.95	18.34	1.59	-3.49
Kazakhstan	France	-61.79	18.34	1.09	-3.49
Kazakhstan	United Kingdom	-60.21	17.60	2.55	-3.49
Kazakhstan	Poland	-58.51	17.39	1.52	-3.49
Russia	Italy	-55.91	15.58	2.20	-18.99
Russia	Germany	-55.77	15.38	2.36	-18.99
Russia	France	-55.73	15.58	1.51	-18.99
Russia	United Kingdom	-53.11	14.56	3.53	-18.99
Ukraine	Italy	-52.66	14.72	3.65	-9.02
Ukraine	France	-52.47	14.73	2.51	-9.02
Ukraine	Germany	-52.16	14.39	3.91	-9.02
Russia	Poland	-50.87	14.26	2.11	-18.99
Ukraine	United Kingdom	-47.68	13.04	5.85	-9.02
Ukraine	Poland	-44.55	12.55	3.49	-9.02

Notes: Impact of CBAM on bilateral trade flows between countries with major emission reductions and EU importers. Trade share reduction represents the percentage change in π_l^{ij}.

Table 11: Chemical Sector: High Emission and Trade Share Reduction Exporters

Exporter	Importer	Trade Share Reduction (%)	Tariff Rate (%)	Price Ratio	Emission Reduction (kton)
Iran	Italy	-67.13	24.15	6.96	-6.97
Iran	France	-67.10	23.99	6.96	-6.97
Algeria	Italy	-66.08	23.58	6.96	-2.84
Algeria	France	-66.05	23.42	6.96	-2.84
Russia	Italy	-65.36	23.28	6.96	-23.02
Russia	France	-65.33	23.12	6.96	-23.02
Iran	United Kingdom	-65.32	22.88	6.96	-6.97
Iran	Germany	-64.70	22.51	6.05	-6.97
Algeria	United Kingdom	-64.20	22.31	6.96	-2.84
Russia	United Kingdom	-63.43	22.00	6.96	-23.02
Iran	Poland	-63.28	22.30	3.45	-6.97
Algeria	Germany	-62.11	21.10	6.96	-2.84
Russia	Germany	-61.30	20.79	6.96	-23.02
Russia	Poland	-58.58	19.97	4.30	-23.02
Qatar	Italy	-57.99	18.72	6.96	-1.04

Notes: Impact of CBAM on bilateral trade flows between countries with major emission reductions and EU importers. Trade share reduction represents the percentage change in π_l^{ij}.

Overall, these results point to the success of CBAM in combination with lower free allocation in terms of carbon leakage protection and European competitiveness. This first indication must be tested, compared, and verified in future research.

8 Conclusion

This thesis contributes to the quantitative analysis of EU climate policy by extending a comprehensive structural gravity model that now captures the complex interactions between carbon pricing, free allocation, and carbon border adjustment. The model represents a significant methodological advancement in the economic modeling of carbon leakage protection instruments, particularly the transition and interaction between free allocation and CBAM within the EU-ETS framework.

The core contribution lies in three advancements in the model. First, the integration of a free allocation subsidy that reflects actual EU-ETS allocation rules and incorporates empirical data from the European Union Transaction Log. Second, the adaptation of existing carbon tariff equations to account for free allocation during the transition period, creating a systematic interaction between both policy instruments. Third, the refinement of sectoral coverage definitions for both EU-ETS and CBAM sectors, enhancing the precision of policy impact analysis.

Another contribution involves data analysis and preparation needed for the new model features. This improves realism using real-world EU-ETS data and accuracy by introducing the necessary correction factor. The added variables based on this data are key to combine the quantity-based EU-ETS allocation system with the cost share structure required for general equilibrium analysis.

The modeling approach using sequential shocks represents another important innovation. Using two distinct policy stages enables the isolation of the CBAM-specific effects from broader carbon pricing impacts and is a pragmatic solution to the data limitations regarding world input-output tables.

While preliminary results suggest surprisingly efficient carbon leakage protection and improvements in European competitiveness, these findings require validation through comparison scenarios and robustness testing in future research.

The modeling framework developed in this thesis establishes a foundation for ongoing

policy analysis as EU climate policy continues to evolve. The planned expansion of CBAM coverage or complete phase-out of free allocation can be captured by the current model structure. As other countries consider similar border adjustment mechanisms, this methodological approach could also be used for policy impact assessment beyond the EU context.

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A Proof Free Allocation and CBAM equivalence

In the following I will show that the value of free allowances allocated to some EU-producer and the CBAM tariff paid are the same when using the official EU's formula or the proposed model. I omit the fade out factor for simplicity.

Assume an EU producer with emission intensity $I^{EU} = a\frac{t\ CO2}{t\ good}$ and a US producer with $I^{US} = b\frac{t\ CO2}{t\ good}$. These intensities could be scaled with some unknown country specific price per ton $\bar{p}^{EU}$ and $\bar{p}^{US}$ given in $\frac{€}{t}$. These prices are not the factory-gate prices p_l^i in the model because those do not refer to money per physical unit which is needed here to scale intensities per physical unit to intensities per monetary unit.

The emission intensities per value of the good are subsequently $\iota^{EU} = \alpha\frac{t\ CO2}{€} = a \cdot \hat{p}_{EU}^{-1}$ and $\iota^{US} = \beta\frac{t\ CO2}{€} = b \cdot \hat{p}_{US}^{-1}$.

The EU benchmark m in $\frac{t\ CO2}{t\ good}$ defines free allocation. The actual intensity of the EU producer a and m define the empirical free allocation share $s^{EU} = \frac{m}{a}\frac{\frac{t\ CO2}{t\ good}}{\frac{t\ CO2}{t\ good}} = \frac{m}{a}$.

The firms produce the volume V^{US} (for the European market) and V^{EU} in tons. The volumes translate into production and import values with the above mentioned unknown price per ton $\bar{p}^{EU}$ and $\bar{p}^{US}$ in $\frac{€}{t}$. This leads to import values $V^{EU} \cdot \bar{p}^{EU} = Y^{EU}$ and $V^{US} \cdot \bar{p}^{US} = X^{US}$

Using the EU formulas for value of free allocation (VFA) and CBAM payment, where P^{EUA} is the price of a certificate yields:

$$VFA = m \cdot V^{EU} \cdot P^{EUA}$$

$$CBAM = (b - m) \cdot V^{US} \cdot P^{EUA}$$

This can be transformed as follows:

$$VFA = m \cdot V^{EU} \cdot P^{EUA} = \frac{a}{a} \cdot m \cdot V^{EU} \cdot \frac{\bar{p}^{EU}}{\bar{p}^{EU}} \cdot P^{EUA} = \frac{a}{\bar{p}^{EU}} \cdot \frac{m}{a} \cdot V^{EU} \cdot \bar{p}^{EU} = \alpha \cdot s^{EU} \cdot Y^{EU} \cdot P^{EUA}$$

$$CBAM = (b - m) \cdot V^{US} \cdot P^{EUA} = (b - m \cdot \frac{a}{a} \cdot \frac{\bar{p}^{US}}{\bar{p}^{EU}} \cdot \frac{p^{EU}}{p^{US}}) \cdot V^{US} \cdot P^{EUA}$$

$$= (\frac{b}{\bar{p}^{US}} - \frac{a}{\bar{p}^{EU}} \cdot \frac{m}{a} \cdot \frac{p^{EU}}{p^{US}}) \cdot V^{US} \cdot \bar{p}^{US} \cdot P^{EUA}$$

$$= (\beta - \alpha \cdot s^{EU} \cdot \frac{p^{EU}}{p^{US}}) \cdot X^{US} \cdot P^{EUA}$$

Although the unknown prices and the factory gate prices are unequal ($\bar{p}^{US} \neq p^{US}$ and $\bar{p}^{EU} \neq p^{EU}$) because $\bar{p}$ is defined relative to physical units of the good, their ratios $\frac{\bar{p}^{US}}{\bar{p}^{EU}}$ and $\frac{p^{US}}{p^{EU}}$ are the same as those are price ratios independent of the unit they refer to.

In the model I eventually use exogenous prices anyway due to the lack of values for the factory gate prices. However, abstracting from this data constraint, using the prices endogenous to the model would be the exact solution.

α and β are defined as emission intensity per product value and can be expressed as $\alpha = \frac{\xi_R^{EU} \alpha_{lE}^{EU}}{(1+\bar{\lambda}_l^{EU})\bar{r}}$ and $\beta = \frac{\xi_R^{US} \alpha_{lE}^{US}}{r}$. In the language of the model $P^{EUA} = \lambda_l^{EU} r$. The price ratio $\frac{p^{EU}}{p^{US}}$ is equal to the price correction factor $p_l^{EU,US}$ in the model (which I then estimate using exogenous prices).

Using these definitions and dividing by the production and export values Y^{EU} and X^{US} to obtain lump-sum tax rates yields the exact definitions of the free allocation subsidy and the CBAM tariff rate of the model (without fade out factor):

$$\chi_l^i - 1 = \frac{\xi_R^i \alpha_{lE}^i}{(1 + \bar{\lambda}_l^i)\bar{r}} \cdot s_l^i \cdot \lambda_l^i r$$

$$t_l^{ij} = \left(\frac{\xi_R^i \alpha_{lE}^i}{r} - \frac{\xi_R^j \alpha_{lE}^j}{(1 + \bar{\lambda}_l^j)\bar{r}} \cdot s_l^j \cdot \bar{p}_l^{ji} \right) \cdot \lambda_l^j r$$

B Examples Price Correction Factor

In this section, I sketch two examples to show why the price correction factor is necessary. The consequences of omitting the correction factors become visible when comparing the amount of emissions that are not covered by free allocation in the EU and the amount of emissions priced under CBAM for the same quantity of the good. I obtain the amount of emissions by multiplying the covered intensity by the price (product value).

In the first example, India has lower factor prices and can produce cheaper compared

Table 12: Data Example I: Lower Factor Prices, Higher Emissions

Country	Product Price (per ton)	Emission Intensity	Emissions (per ton)	Free Allocation Share
France	$p^F = 40$ €	$z^F = 0.1$	$0.1 \cdot 40 = 4\ t\ CO_2$	$s^F = 1$
India	$p^I = 20$ €	$z^I = 0.5$	$0.5 \cdot 20 = 10\ t\ CO_2$	–

Note: These numbers are fictitious and only serve illustrative purposes.

to France. But India also has higher emissions per ton of output and exports the good to France. The French company faces a free allocation share of $s^F = 1$ and is fully reimbursed for the carbon tax it paid. The entire 4 tons of CO_2 are covered by free allocation. Hence, the company from India should pay the tariff only for $10 - 4 = 6$ tons of CO_2 correspondingly.

Directly comparing the emission intensities in the CBAM formula would leave an intensity of $0.5 - 0.1$ covered by CBAM. Expressed in emission levels, the Indian exporter is charged a tariff for $(0.5 - 0.1) \cdot 20 = 8$ t CO_2. If the comparison includes a price adjustment factor this changes to $(0.5 - 0.1 \cdot \frac{40}{20}) \cdot 20 = 0.3 \cdot 20 = 6$, which is precisely the 6 tons of CO_2 which should be priced.

Table 13: Data Example II: Higher Factor Prices, Lower Emissions

Country	Product Price (per ton)	Emission Intensity	Emissions (per ton)	Free Allocation Share
Korea	$p^K = 40$ €	$z^K = 0.1$	$0.1 \cdot 40 = 4\ t\ CO_2$	–
Romania	$p^R = 20$ €	$z^R = 0.4$	$0.4 \cdot 20 = 8\ t\ CO_2$	$s^R = 0.5$

Note: These numbers are fictitious and only serve illustrative purposes.

In the second example, the factor prices in the exporting country Korea are higher compared to the importing EU country Romania. But production in Korea is already less emission intensive. Only half of the Romanian company's emissions (4 tons of CO_2) are covered by free allocation with $s^R = 0.5$. Subsequently, there should be no tariff applied to the imports from Korea.

Yet, comparing the emission intensities directly would lead to a negative amount of emissions covered by the tariff, i.e. a subsidy for the Korean exporter: $(0.1 - 0.4 \cdot 0.5) \cdot 40 = -4\ t\ CO_2$ Using the price correction factor leads to the correct result, where no emissions are subject to the tariff: $(0.1 - 0.4 \cdot 0.5 \cdot \frac{20}{40}) \cdot 40 = 0$.

C Assessment of non-EU Carbon Pricing

Using data from the State and Trends of Carbon Pricing Dashboard (World Bank Group, 2023) from 1 April 2024 I first filter the data on the "Compliance_Gen Info" sheet following these steps:

In the column "Type", I exclude subnational pricing systems.

In the column "Status", I exclude pricing systems under consideration and under development or pricing systems abolished.

In the column "Jurisdiction covered", I exclude countries of the EEA and Switzerland. Finally, I exclude Albania, due to missing information in the dataset.

The original dataset as well as the remaining observations after this filtering can be found in the replication files.

The following table 14 lists the country, its pricing system, and the reason I classified it as not equivalent to EU-ETS standards. I exclude the UK from the table, because its pricing system is very similar to EU-ETS. The only remaining country with a comparable pricing system is Singapore. I decided to neglect this single pricing systems to reduce complexity of the model and stay within the scope of this thesis.

Free allocation and grandfathering or benchmark systems are not substantially different to the EU-ETS. But given that CBAM is specifically designed to adjust for the amount of free allocation, an adjustment for domestic carbon prices would need to reflect domestic free allocation either. As this is not possible within the scope of this thesis, free allocation and similar systems are excluded.

Table 14: Assessment of non-EU Carbon Pricing

Instrument Name	**Type**	**Reason**	**Reference Column Dataset**
New Zealand ETS	National ETS	Free allocation with a marginal phase-out rate of 1%; Auctions introduced in 2021 but none of them cleared	Pricing and Allocation; Recent Development

Continued on next page

Table 14 – *Continued from previous page*

Instrument Name	**Type**	**Reason**	**Reference Column Dataset**
Japan carbon tax	National Carbon tax	Low price at 1.91 USD	Price on 1 April
Kazakhstan ETS	National ETS	Low price at 1.06 USD; Unlimited offsets	Price on 1 April; Offset eligibility
Mexico carbon tax	National Carbon tax	Unlimited offsets; Only emissions above the natural gas benchmark covered	Offset eligibility; Description
Korea ETS	National ETS	Free allowances for EITE sectors; Auction for at least 10%; Offsets allowed	Description; Offset eligibility
Australia Safeguard Mechanism	National ETS	Unlimited offsets; Only emissions above a benchmark covered	Offset eligibility; Allocation
South Africa carbon tax	National Carbon tax	Tax exemptions of 60–95%; Offsets allowed	Coverage; Offset eligibility
Chile carbon tax	National Carbon tax	Unlimited offsets, low price at 5 USD	Offset eligibility; Price on 1 April
Colombia carbon tax	National Carbon tax	No coverage of coal or natural gas; Unlimited offsets	Fuels covered; Offset eligibility
Ukraine carbon tax	National Carbon tax	Low price at 0.77 USD	Price on 1 April
Canada federal OBPS	National ETS	Only covers emissions above the national average intensity and very large emitters	Coverage; Pricing and allocation
China national ETS	National ETS	Free allocation based on benchmarks; Only covers electricity and heat	Description; Electricity and heat

Continued on next page

Table 14 – *Continued from previous page*

Instrument Name	**Type**	**Reason**	**Reference Column Dataset**
Mexico pilot ETS	National ETS	Grandfathering; Offsets allowed	Allocation; Offset eligibility
Canada federal fuel charge	National Carbon tax	Temporarily paused for a three year period starting October 2023	Recent development
Argentina carbon tax	National Carbon tax	Low price at 0.8 USD	Price on 1 April
Indonesia ETS	National ETS	Only covers a selection of coal-fired power plants	Description
Montenegro ETS	National ETS	Covers only 3 plants	Description
Uruguay CO_2 tax	National Carbon tax	Only applies for gasoline	Fuels covered

D Additional Results First Shock

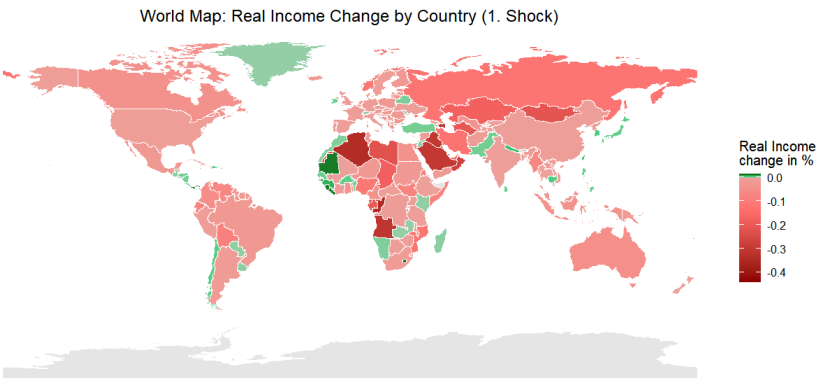


Figure D1: World Map: Real Income Change by Country (1. Shock)

Notes: The color scale turns from red to green at 0.

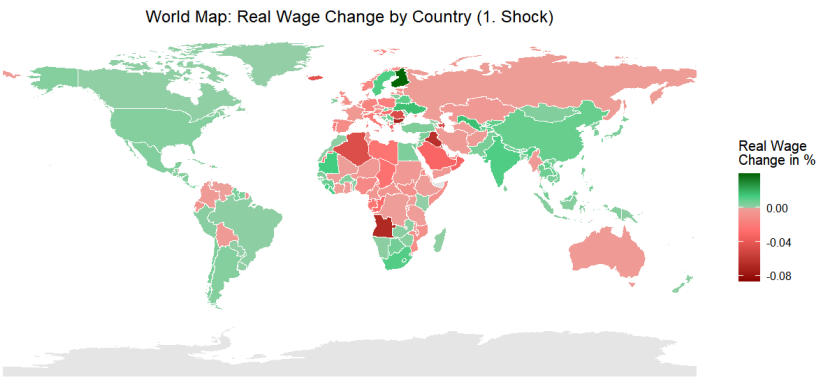


Figure D2: World Map: Real Wage Change by Country (1. Shock)

Notes: The color scale turns from red to green at 0.

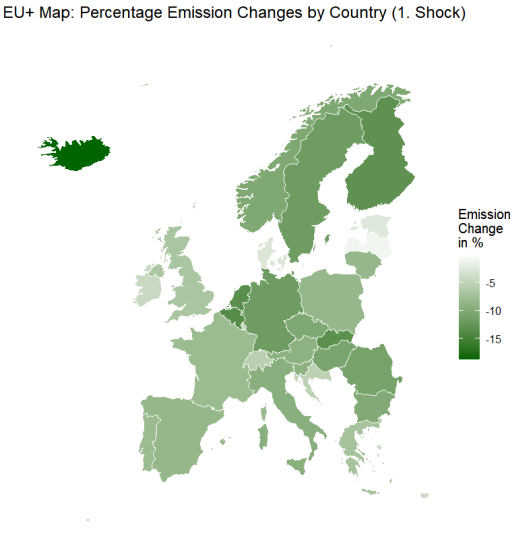


Figure D3: EU+ Map: Percentage Emission Change by Country (1. Shock)

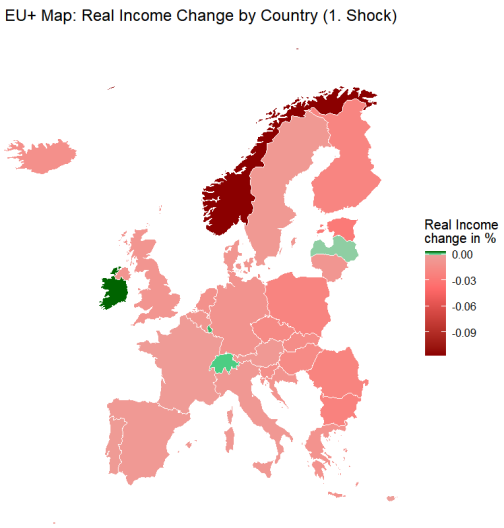


Figure D4: EU+ Map: Real Income Change by Country (1. Shock)

Notes: The color scale turns from red to green at 0.

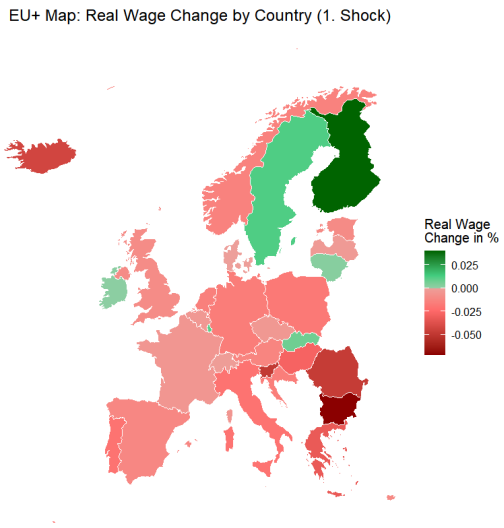


Figure D5: EU+ Map: Real Wage Change by Country (1. Shock)

Notes: The color scale turns from red to green at 0.

Table D1: EU Market Share Changes by Sector (1. Shock)

	Agriculture	Apparel	Chemical	Equipment	Food	Machinery
Change in %	0.04	-0.07	0.23	-0.07	0.03	-0.16

	Metal	Mineral	Other	Paper	Service	Textile	Wood
Change in %	0.03	0.40	-0.08	0.41	0.04	-0.03	0.00

Notes: Change in global market shares by sector of the aggregate EU region expressed as percentage points.

Table D2: Subsidy Rate by Country and Sector (1. Shock)

Country	Chemical	Mineral	Metal	Paper
Germany	0.93%	0.55%	0.35%	0.63%
Spain	0.5%	0.84%	0.74%	0.32%
France	0.6%	0.5%	0.41%	0.42%
Italy	0.58%	0.55%	0.28%	0.36%
Poland	1.7%	1.37%	0.92%	0.72%

Notes: Selection of countries only for illustrative purposes.

E Additional Results Second Shock

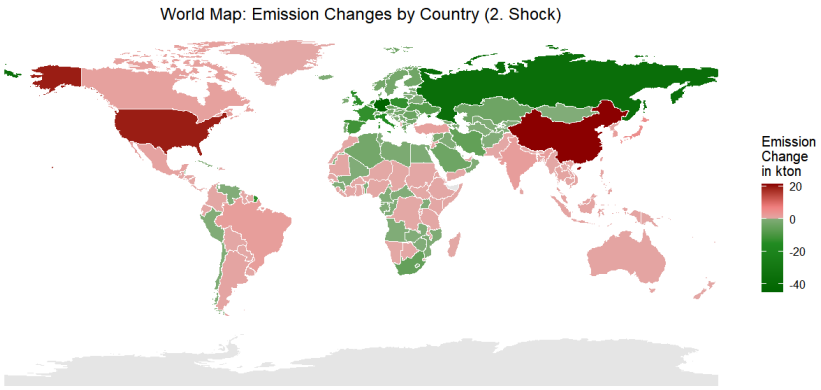


Figure E1: World Map: Absolute Emission Reduction by Country (2. Shock)

Notes: The color scale turns from green to red at 0.

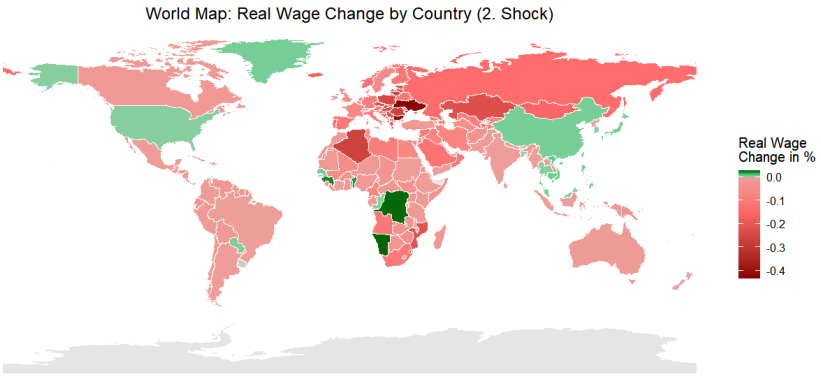


Figure E2: World Map: Real Wage Change by Country (2. Shock)

Notes: The color scale turns from red to green at 0.

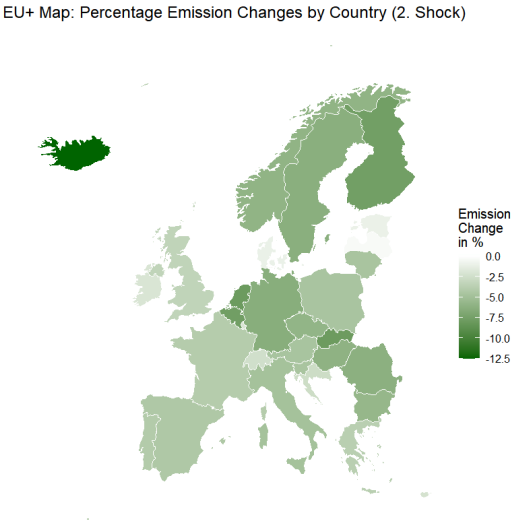


Figure E3: EU+ Map: Percentage Emission Change by Country (2. Shock)

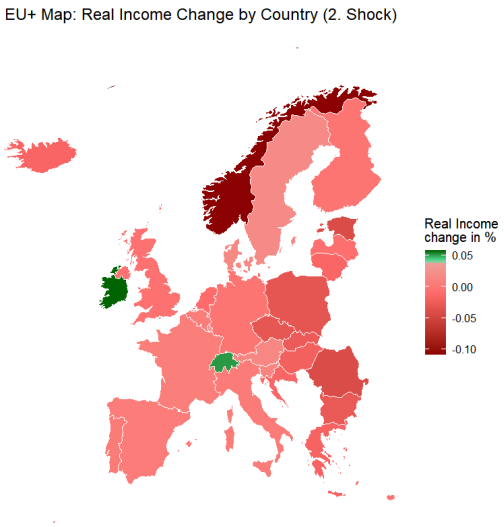


Figure E4: EU+ Map: Real Income Change by Country (2. Shock)

Notes: The color scale turns from red to green at 0.

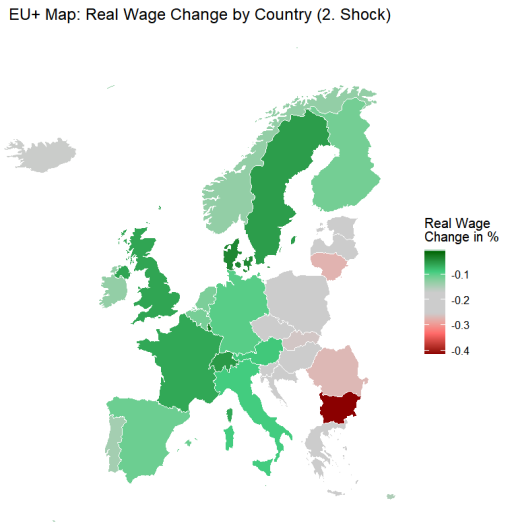


Figure E5: EU+ Map: Real Wage Change by Country (2. Shock)

Notes: The color scale turns from red to green at 0.

Table E1: Change in Subsidy Rate by Country and Sector (2. Shock)

Country	Chemical	Mineral	Metal	Paper
Germany	−0.23%	−0.13%	−0.09%	0.32%
Spain	−0.12%	−0.2%	−0.18%	0.16%
France	−0.15%	−0.12%	−0.1%	0.22%
Italy	−0.14%	−0.13%	−0.07%	0.18%
Poland	−0.41%	−0.33%	−0.22%	0.36%

Notes: Shows the change in the subsidy rate compare to the equilibrium after the 1. shock. Subsidy rates for CBAM sectors decrease due to fade out but increase in the paper sector due to the higher tax rate. Selection of countries only for illustrative purposes.

Table E2: France: Imposed Tariff by Exporter and Sector

Sector	Brazil	China	India	South Korea	Saudi Arabia	Turkey	United States
Chemical	0.33%	4.99%	5.58%	4.13%	12.24%	1.09%	3.14%
Mineral	0%	4.51%	10.89%	1.51%	4.71%	5.72%	1.92%
Metal	2.92%	7.74%	10.78%	1.41%	3%	2.87%	1.79%

Notes: Tariff rate imposed by France to non-EU countries. Selection of trading partners only for illustrative purposes.

Table E3: Mineral: High Emission and Trade Share Reduction Exporters (extensive)

Exporter	Importer	Trade Share Reduction (%)	Tax Rate (%)	Price Ratio	Emission Reduction (kton)
Chile	Germany	-13.50	7.86	0.70	-0.58
Algeria	Germany	-13.98	8.48	0.76	-0.29
Russia	Germany	-10.58	6.84	0.94	-0.39
Tunisia	Germany	-27.56	16.35	3.40	-0.37
Turkey	Germany	-9.54	5.72	1.84	-0.31
South Africa	Germany	-12.79	7.93	2.57	-0.56
Chile	France	-13.29	7.86	0.76	-0.58
Algeria	France	-13.77	8.48	0.83	-0.29
Russia	France	-10.36	6.84	1.03	-0.39
Tunisia	France	-27.40	16.36	3.71	-0.37
Turkey	France	-9.33	5.72	2.00	-0.31
South Africa	France	-12.59	7.94	2.80	-0.56
Chile	United Kingdom	-12.95	7.68	0.84	-0.58
Algeria	United Kingdom	-13.40	8.28	0.91	-0.29
Russia	United Kingdom	-9.87	6.59	1.13	-0.39
Tunisia	United Kingdom	-26.04	15.45	4.08	-0.37
Turkey	United Kingdom	-8.33	5.23	2.20	-0.31
South Africa	United Kingdom	-11.26	7.25	3.08	-0.56
Chile	Italy	-13.15	7.73	0.94	-0.58
Algeria	Italy	-13.61	8.33	1.02	-0.29
Russia	Italy	-10.13	6.66	1.26	-0.39
Tunisia	Italy	-26.53	15.71	4.55	-0.37
Turkey	Italy	-8.74	5.37	2.46	-0.31
South Africa	Italy	-11.76	7.45	3.44	-0.56
Chile	Poland	-10.87	7.23	0.74	-0.58
Algeria	Poland	-11.26	7.79	0.81	-0.29
Russia	Poland	-7.39	5.99	1.00	-0.39
Tunisia	Poland	-21.56	13.27	3.59	-0.37
Turkey	Poland	-4.56	4.05	1.94	-0.31
South Africa	Poland	-6.68	5.61	2.72	-0.56

Notes: Impact of CBAM on bilateral trade flows between countries with major emission reductions and EU importers. Trade share reduction represents the percentage change in π_l^{ij}.

Table E4: Metal: High Emission and Trade Share Reduction Exporters (extensive)

Exporter	Importer	Trade Share Reduction (%)	Tax Rate (%)	Price Ratio	Emission Reduction (kton)
China	Germany	-32.99	7.62	1.34	-0.88
Egypt	Germany	-32.83	7.75	1.22	-0.81
India	Germany	-43.63	10.61	1.93	-3.19
Kazakhstan	Germany	-61.95	18.20	1.70	-3.49
Russia	Germany	-55.77	15.38	2.36	-18.99
Ukraine	Germany	-52.16	14.39	3.91	-9.02
South Africa	Germany	-37.26	9.19	0.64	-5.88
China	France	-32.62	7.74	0.86	-0.88
Egypt	France	-32.42	7.86	0.78	-0.81
India	France	-43.49	10.78	1.24	-3.19
Kazakhstan	France	-61.79	18.34	1.09	-3.49
Russia	France	-55.73	15.58	1.51	-18.99
Ukraine	France	-52.47	14.73	2.51	-9.02
South Africa	France	-36.69	9.25	0.41	-5.88
China	United Kingdom	-30.23	7.16	2.00	-0.88
Egypt	United Kingdom	-30.25	7.33	1.82	-0.81
India	United Kingdom	-40.64	9.94	2.89	-3.19
Kazakhstan	United Kingdom	-60.21	17.60	2.55	-3.49
Russia	United Kingdom	-53.11	14.56	3.53	-18.99
Ukraine	United Kingdom	-47.68	13.04	5.85	-9.02
South Africa	United Kingdom	-35.65	8.97	0.95	-5.88
China	Italy	-32.90	7.74	1.25	-0.88
Egypt	Italy	-32.71	7.86	1.14	-0.81
India	Italy	-43.72	10.77	1.80	-3.19
Kazakhstan	Italy	-61.95	18.34	1.59	-3.49
Russia	Italy	-55.91	15.58	2.20	-18.99
Ukraine	Italy	-52.66	14.72	3.65	-9.02
South Africa	Italy	-36.96	9.25	0.59	-5.88
China	Poland	-27.37	6.99	1.20	-0.88
Egypt	Poland	-27.45	7.18	1.09	-0.81
India	Poland	-37.95	9.70	1.73	-3.19
Kazakhstan	Poland	-58.51	17.39	1.52	-3.49
Russia	Poland	-50.87	14.26	2.11	-18.99
Ukraine	Poland	-44.55	12.55	3.49	-9.02
South Africa	Poland	-33.37	8.89	0.57	-5.88

Notes: Impact of CBAM on bilateral trade flows between countries with major emission reductions and EU importers. Trade share reduction represents the percentage change in π_l^{ij}.

Table E5: Chemical: High Emission and Trade Share Reduction Exporters (extensive)

Exporter	Importer	Trade Share Reduction (%)	Tax Rate (%)	Price Ratio	Emission Reduction (kton)
Algeria	Germany	-62.11	21.10	6.96	-2.84
Iran	Germany	-64.70	22.51	6.05	-6.97
Qatar	Germany	-54.04	16.76	6.39	-1.04
Russia	Germany	-61.30	20.79	6.96	-23.02
Saudi Arabia	Germany	-39.13	11.17	4.37	-5.07
Serbia	Germany	-42.05	12.29	2.52	-1.42
Algeria	France	-66.05	23.42	6.96	-2.84
Iran	France	-67.10	23.99	6.96	-6.97
Qatar	France	-57.95	18.56	6.96	-1.04
Russia	France	-65.33	23.12	6.96	-23.02
Saudi Arabia	France	-42.49	12.24	5.02	-5.07
Serbia	France	-43.99	12.91	2.89	-1.42
Algeria	United Kingdom	-64.20	22.31	6.96	-2.84
Iran	United Kingdom	-65.32	22.88	6.96	-6.97
Qatar	United Kingdom	-55.56	17.44	6.96	-1.04
Russia	United Kingdom	-63.43	22.00	6.96	-23.02
Saudi Arabia	United Kingdom	-39.05	11.13	5.43	-5.07
Serbia	United Kingdom	-42.04	12.27	3.13	-1.42
Algeria	Italy	-66.08	23.58	6.96	-2.84
Iran	Italy	-67.13	24.15	6.96	-6.97
Qatar	Italy	-57.99	18.72	6.96	-1.04
Russia	Italy	-65.36	23.28	6.96	-23.02
Saudi Arabia	Italy	-41.89	12.16	5.36	-5.07
Serbia	Italy	-43.50	12.86	3.09	-1.42
Algeria	Poland	-53.43	17.34	6.03	-2.84
Iran	Poland	-63.28	22.30	3.45	-6.97
Qatar	Poland	-52.15	16.54	3.64	-1.04
Russia	Poland	-58.58	19.97	4.30	-23.02
Saudi Arabia	Poland	-36.82	11.02	2.49	-5.07
Serbia	Poland	-40.05	12.21	1.43	-1.42

Notes: Impact of CBAM on bilateral trade flows between countries with major emission reductions and EU importers. Trade share reduction represents the percentage change in π_l^{ij}.